Bayesian Structural Inference for Dynamic Crowdsourcing Contests

Jussi Keppo
NUS Business School and Institute of Operations Research and Analytics
National University of Singapore, Singapore
keppo@nus.edu.sg

Linsheng Zhuang
Institute of Operations Research and Analytics
National University of Singapore, Singapore
linsheng.z@u.nus.edu

Online crowdsourcing contests generate rich dynamic behaviour, but their core strategic state is latent: contestants react to noisy public leaderboard feedback rather than true relative performance. We develop a tractable dynamic game with Bayesian filtering and a corresponding structural estimation framework to recover latent competition state, effort dynamics, participant ability, innovation volatility, and signal precision using only contest-phase information available to participants. The model admits a closed-form Markov Perfect Equilibrium, enabling disciplined Bayesian estimation in a high-frequency dynamic setting. Our key empirical design feature is an external validation design based on the private leaderboard, which is excluded from estimation, not observed by contestants during the contest, and nonetheless generated by the same underlying process. Applying the framework to 73 Kaggle contests, we find statistically meaningful but uneven alignment between structurally recovered latent quantities and private-leaderboard benchmarks constructed from excluded post-contest information. The evidence is stronger for innovation risk than for signal precision, supporting credible partial validation and providing a practical framework for analyzing and designing digital contest environments.

Key words: Bayesian Statistics, Structural Estimation, Crowdsourcing Contest, Dynamic Contest

1. Introduction

Crowdsourcing contests are widely used by firms to source solutions to complex problems from external contributors. Platforms such as Kaggle[1], Topcoder[2], DrivenData[3], and AIcrowd[4] host thousands of these contests, attracting data scientists, engineers, and developers to compete by submitting solutions to real-world problems. On Kaggle, for instance, a typical contest spans several weeks, during which participants compete for monetary prizes by repeatedly submitting solutions, observing their relative ranking on a public leaderboard, and adjusting their strategies in response to evolving feedback and the observed actions of competitors.

The central empirical challenge in these online crowdsourcing contests is that the true competitive state is unobserved during the contest. Participants observe noisy public leaderboard updates rather than true relative performance, and they choose effort based on filtered beliefs rather than on the latent state itself. The econometrician faces the same partial-observability problem, making it difficult to recover strategic behaviour and incentives from contest data.

Motivated by this identification problem, we develop a dynamic game-theoretic model of contests with noisy public signals and Bayesian belief updating, and we take it to data using a structural Bayesian framework. The model is designed for settings such as Kaggle, where effort is continuous, strategic interaction is dynamic, and observed leaderboard information is only a noisy projection of underlying performance. Using Meta-Kaggle data (Risdal and Bozsolik 2022), we infer latent strategic objects from contest-phase data and then evaluate the resulting estimates with withheld post-contest information.

To illustrate the framework in greater detail, Figure 1 depicts the key components and their interactions in a typical Kaggle contest. One notable feature is the existence of two leaderboards: a *private* leaderboard, which is hidden until the contest ends and ultimately determines the final ranking and prize allocation, and a *public* leaderboard, which is visible to all participants during the contest.[5] Both leaderboards are updated upon each submission. After observing the public leaderboard, contestants update their beliefs and form a perceived state of the competition. Based on this belief, they choose their effort level, which corresponds to a *Markov Perfect Equilibrium* (MPE). These effort levels shape the submission intensity, and each submission triggers updates to

[1] Kaggle: https://www.kaggle.com

[2] Topcoder: https://www.topcoder.com

[3] DrivenData: https://www.drivendata.org

[4] AIcrowd: https://www.aicrowd.com

[5] This design serves two purposes. First, the public leaderboard provides participants with real-time feedback. Second, by keeping the private leaderboard concealed, the platform discourages overfitting to the public metric and preserves uncertainty, thereby sustaining competitive tension throughout the contest.

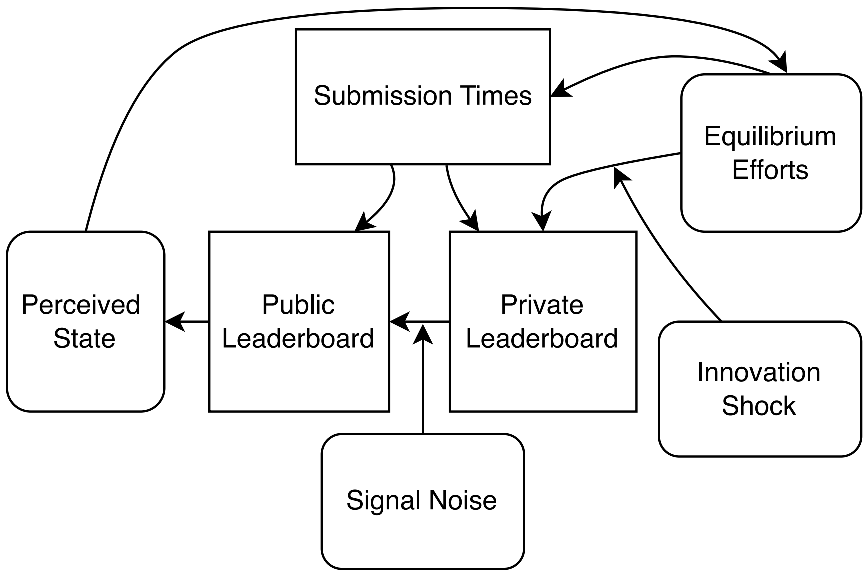


Figure 1 Structure of a Typical Kaggle Contest

Note: This figure illustrates the structure of a typical Kaggle contest and the information flow available to contestants. Rectangles and rounded rectangles represent observed data and latent states, respectively. The private leaderboard is visible only to the contest organizer during the contest and is revealed to all participants only after it ends. During the contest period, contestants can observe only the submission times of each player and the public leaderboard.

both leaderboards. The private leaderboard, in particular, evolves according to the real-time effort level and an innovation shock.

We estimate the structural parameters of the model using a Bayesian approach, based on players' submission times and the observed *public* leaderboard trajectories from the Meta-Kaggle database. The estimation relies solely on information available to participants during the contest and leverages the equilibrium structure implied by the dynamic model. This procedure enables us to recover a range of latent components that are not directly observable in the data, including the perceived state of the competition, the evolution of effort over time, individual ability levels, innovation uncertainty, and the precision of public signals.

To assess whether the estimated model captures the underlying structure of real-world contests, we perform a cross-validation exercise using out-of-sample data from the *private* leaderboard. As illustrated in Figure 1, the volatility of the private leaderboard reflects the level of innovation risk in each contest, while the discrepancy between the private and public leaderboards captures the precision of the observed signal. We estimate these contest-specific benchmark quantities and compare them against the corresponding values inferred from the Bayesian structural model. This mapping is institutionally grounded: both leaderboards are generated from the same submissions and scoring process, but only the public leaderboard is observed during the contest and is typically computed on a subset of the evaluation data. Accordingly, private-leaderboard volatility provides a natural proxy for underlying innovation uncertainty, while public-private divergence provides a benchmark for signal informativeness. We use this mapping as a disciplined validation target rather than as an additional estimation input.

Empirically, we carefully select 73 high-quality contests from the Meta-Kaggle database and preprocess the data to ensure consistency across estimation methods. For each contest, we obtain two sets of contest-specific parameters—innovation risk and signal precision—using (i) our Bayesian structural estimation and (ii) a reduced-form approach based on private leaderboard dynamics. We then perform regression analysis to compare the results from the two methods. The findings show a positive and statistically significant correlation between the two sets of estimates, with stronger recovery for innovation risk and noisier recovery for signal precision.

This paper makes two main contributions. First, we develop a dynamic game-theoretic model and a Bayesian structural framework for contests with noisy public feedback and latent performance dynamics. Relative to contest models centered on discrete submit-or-stop decisions, our approach models continuous effort accumulation and makes belief updating from noisy signals structurally central. Building on the tractable equilibrium structure in Ryvkin (2022), we introduce partial observability in a way that remains empirically estimable and behaviourally interpretable. Second, we propose a validation design that exploits the private leaderboard as a naturally excluded data source: it is not available to contestants during the contest and is not used in estimation, yet it is generated by the same underlying process. This creates a disciplined test of whether the estimated structure recovers innovation uncertainty and signal precision rather than merely fitting observed contest-phase behaviour. The results provide credible partial validation of the model and highlight its value for mechanism evaluation and platform design. Taken together, these contributions advance our understanding of online competitive dynamics and provide a foundation for future empirical work on digital contest design.

1.1. Related Literature

This study is most closely related to the following strands of literature.

Contest Design. A rich literature on *static* contest design (Ales et al. 2017) investigates how to structure competitive environments to elicit performance, typically through a contest success function mapping effort to winning probabilities (Skaperdas 1996).[6] Because our setting is dynamic with repeated feedback, this static branch serves mainly as background for the mechanism-design questions we build on, including prize allocation, participation rules, and information disclosure (Moldovanu and Sela 2001, Ales et al. 2021, Bergemann et al. 2022).

[6] There are exceptions like beauty or barrier contests.

While these studies offer valuable insights into incentive design, they largely overlook how participants dynamically adjust their efforts in response to evolving competition. In contrast, this paper focuses on *dynamic* contests, where strategic decisions unfold over time. Two main modeling approaches have been used to capture such dynamic strategic interactions:

One strand of the literature models dynamic contests as multistage processes, where participants exert effort over time while facing uncertainty about intermediate progress. This structure facilitates the analysis of the mid-contest policy instruments available to the contest designer, such as feedback disclosure and screening mechanisms. Specifically, in a typical two-stage setting (Aoyagi 2010, Mihm and Schlapp 2019), the initial stage captures either feasibility uncertainty or foundational experimentation, with progress revealed through milestone feedback provided by the contest organizer. Participants exert effort in both stages, and the final output reflects the cumulative impact of early performance and subsequent improvement. Similarly, Bimpikis et al. (2019) study a two-stage model in which agents exert continuous effort to complete Stage A (potentially infeasible) and Stage B (feasible), with breakthroughs governed by Poisson processes. In all these models, the contest designer may strategically disclose interim feedback. Another example is Khorasani et al. (2023), who analyze a two-stage contest in which solvers invest exploratory effort to generate viable ideas, with only those passing an imperfect screening mechanism proceeding to execution.

The second approach, more closely related to our work, treats output as a stochastic process whose drift depends on the participant's effort. This line of research originates from the tug-of-war framework, first formalized by Harris and Vickers (1987) as a one-dimensional simplification of a multi-stage R&D race. Subsequent studies build on this foundation by modeling each player's progress as a Brownian motion. Budd et al. (1993), for example, analyze a duopoly innovation race where the state evolves as a Brownian motion driven by the difference in effort, and characterize an approximate equilibrium. Moscarini and Smith (2011) advance this approach by modeling the game state as the output gap between players and derive closed-form equilibrium strategies under pure-strategy play. Ryvkin (2022) adapt their framework to a dynamic contest with a fixed deadline, providing a closed-form solution for the Markov perfect equilibrium.

This paper studies online crowdsourcing contests, which typically feature fixed deadlines, repeated submissions, and public real-time feedback, all governed by platform-specific rules and evaluation metrics. Hence, our analysis builds on the model by Ryvkin (2022), whose simplifications yield a closed-form equilibrium that makes structural estimation in a dynamic setting analytically and computationally tractable. We depart from their full-information setting by introducing a noisy public signal and Bayesian filtering, thereby bringing partial observability into a tractable dynamic contest model.

Structural Estimation of Contests. Beyond theoretical modelling, a growing body of research employs structural estimation techniques to recover key parameters of contest environments using observed data.

In the structural estimation of static contests, when a contest success function (CSF) is explicitly specified, the primary goal is to estimate its functional form or key parameters (Hwang 2012, Kang 2016, Huang and He 2021). In contrast, for contests without a well-defined or smooth CSF—such as beauty contests—researchers typically rely on equilibrium conditions, most notably the first-order condition. For example, Yoganarasimhan (2016) proposes a two-step semiparametric estimator, resembling the EM algorithm, to recover both unobserved auction heterogeneity and bidder cost distributions. Similarly, in auction-like models (Guerre et al. 2000, Shakhgildyan 2022), nonparametric estimation of the underlying valuation structure often serves as a first step, providing the foundation for subsequent structural estimation.

Most dynamic structural estimation exercises rely on the Markov Perfect Equilibrium (MPE) of a dynamic game. However, when the model is calibrated to closely fit real-world data, the MPE typically lacks a closed-form solution. For example, Jiang et al. (2022) develop a finite-horizon dynamic game with discrete time, state, and action spaces to model creators' strategic behaviour in response to feedback during online logo design contests. Using data from 810 contests, they implement a two-step maximum likelihood estimation (MLE) procedure under the assumption of ex-ante homogeneous contests and players: the first step estimates action-specific state transition probabilities, while the second recovers utility parameters.

Specific to online crowdsourcing environments, Chen et al. (2021) use data from 1,980 contests on TaskCN.com to examine how contest duration affects both the number and quality of participants. Through regression analysis, they show that while longer durations attract more contestants, they tend to reduce the quality of top entrants. Zhang et al. (2019) build a dynamic structural model to study how participants in crowdsourcing contests learn and choose contests over time. Their model shows that competing against superstars accelerates learning by providing more informative feedback. Using Bayesian updating and a hierarchical model, they estimate how individuals trade off monetary rewards, learning opportunities, and participation costs, based on data from Topcoder.

Most closely related to our work, Lemus and Marshall (2021) conduct contest-level structural estimation using Kaggle data. They propose a finite-horizon dynamic game in which, after each submission, players decide whether to continue or exit, essentially solving an optimal stopping problem. Players can only make this decision immediately after completing a submission, which arrives at random intervals and incurs a privately drawn cost. Their estimation proceeds in two stages: the first uses maximum likelihood and the EM algorithm to estimate primitive parameters such as the distribution of player types and scores; the second applies a CCP-based estimator to identify the cost distribution. Our model differs in two key respects: (1) we treat submissions as the outcome of accumulated effort rather than as discrete costly actions; and (2) we explicitly model belief updating from noisy signals as a structural channel that shapes both behaviour and identification, whereas their CCP-based approach relies on a coarser behavioural approximation.

The structure of the paper is as follows: Section 2 introduces our theoretical model of online crowdsourcing contests. Section 3 presents the Bayesian estimation framework. In Section 4, we describe the data-generating process and apply the Bayesian framework to synthetic data. Section 5 validates the theoretical model using real-world data. A replication package (code and implementation details) will be provided with the paper.

2. The Model

Two players, i and j, compete for a prize $\theta > 0$ in a contest. The winner receives the prize and the loser receives zero. The contest starts at time zero. At each time $t \geq 0$, player i chooses an effort level $q_{i,t}$ and incurs quadratic cost $C_i(q_{i,t}) = c_i q_{i,t}^2/2$, where lower c_i corresponds to higher ability. Let y_t denote the *output gap* between players i and j at time t, evolving as

$$dy_t = (q_{i,t} - q_{j,t})dt + \sigma dW_t \tag{1}$$

where W_t is a Brownian motion and $\sigma > 0$ measures the innovation risk. Equation (1) is a reduced-form representation of relative progress: effort shifts the drift of performance differences, rather than serving as a literal production technology for leaderboard scores. When σ is large and the relationship between effort and output is highly uncertain, luck becomes the dominant factor. Conversely, when σ is small and output closely reflects effort, the importance of effort is amplified.

The contest is equipped with a submission system that allows participants to upload algorithms over time and receive public feedback. For tractability, we assume submissions occur whenever participants make intermediate progress. The platform can evaluate true performance, while contestants observe only public feedback generated from that underlying process. At any time $t > 0$, contestants observe a public signal of the latent output gap y_t. The precision of this signal is governed by λ. The signal evolves according to

$$dZ_t = y_t dt + \frac{dB_t}{\sqrt{\lambda}} \tag{2}$$

where B_t is a standard Brownian motion independent of W_t, and λ controls signal precision. A larger λ implies less noise and more informative feedback; in the empirical mapping, this corresponds to a more informative public leaderboard signal about latent relative performance. This continuous-time signal is an idealized analogue of the stream of public leaderboard updates observed after submissions.

The information set of both players at time $t \geq 0$ is $I_t \equiv \{Z_s : 0 \leq s \leq t\}$. Player i estimates the unknown output gap y_t based on I_t. Let $\tilde{y}_t \equiv E(y_t \mid I_t)$ denote the filtered mean output gap, and let $S_t \equiv E[(\tilde{y}_t - y_t)^2 \mid I_t]$ denote the conditional mean-squared filtering error. Under standard linear-Gaussian assumptions, the Kalman-Bucy filter (Chapter 1.2 of Bensoussan 1992; see also Frogerais et al. 2012, Barrau and Bonnabel 2017) implies

$$d\tilde{y}_t = (q_{i,t} - q_{j,t})dt + \lambda S_t(dZ_t - \tilde{y}_t dt) \tag{3}$$

$$\frac{dS_t}{dt} = \sigma^2 - \lambda S_t^2 \tag{4}$$

Hence, the conditional distribution satisfies $y_t \mid I_t \sim \mathcal{N}(\tilde{y}_t, S_t)$, so the posterior state is fully characterized by $(\tilde{y}_t, S_t)$. The steady-state value associated with equation (4) is $\bar{S} \equiv \sigma/\sqrt{\lambda}$. For analytical tractability, we set $S_0 = \bar{S}$, so that $S_t \equiv \bar{S}$ for all t. This simplification collapses the state space to $(\tilde{y}_t, t)$ and yields a closed-form equilibrium characterization, but it rules out time-varying filtering uncertainty and therefore limits comparative statics involving information dynamics. The general solution of equation (4) is reported in Appendix B.1. Figure 2 illustrates convergence of S_t to $\bar{S}$ from arbitrary initial conditions.

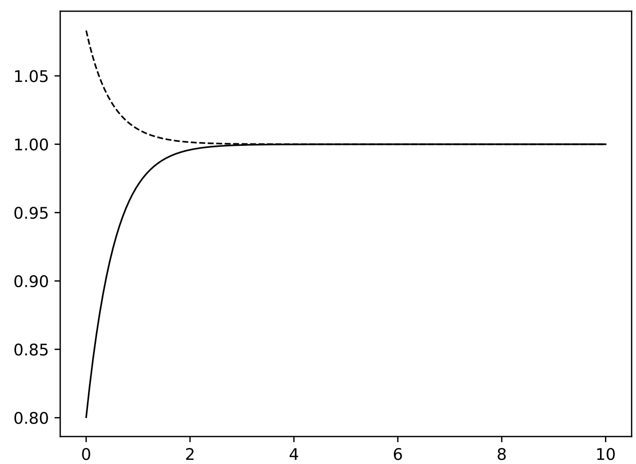


Figure 2 The evolution of S_t in time t, given that $\lambda = 1$ and $\sigma = 1$.

Note: The plot shows the conditional variance S_t of the filtering error (i.e., the uncertainty about the true performance gap). It converges quickly from any initial value to its steady state $\bar{S} = \sigma/\sqrt{\lambda}$. The figure illustrates the convergence of S_t to its steady state.

Following Ryvkin (2022), we consider a dynamic contest with a fixed deadline. Suppose the contest is terminated when time $t = T > 0$. Under the steady-state restriction $S_t \equiv \bar{S}$, the state of the game is fully characterized by $(\tilde{y}_t, t)$. The winner is defined by latent final performance y_T. Under the Gaussian posterior, the winning event can be represented equivalently through a posterior-mean threshold: $\mathbb{P}(y_T > 0 \mid I_T)$ is monotone in $\tilde{y}_T$. We therefore formulate the terminal payoff in terms of $\tilde{y}_T$, which is equivalent under the Gaussian posterior and consistent with Ryvkin (2022). At any time $0 \leq t < T$, player i optimizes her effort level $q_{i,\tau}$ in the remaining contest period $\tau \in [t, T)$ according to the following optimization problem,

$$V^i(\tilde{y}_t, t; q_{j,t}, \Theta_i) = \max_{\{q_{i,\tau}\}_{\tau=t}^T} \mathbb{E}\left(\theta \cdot \mathbf{1}_{\tilde{y}_T > 0} - \int_t^T C_i(q_{i,\tau}) d\tau \middle| I_t\right) \tag{5}$$

where $\Theta_i \equiv \{\theta, \lambda, \sigma, c_i\}$, subject to constraints (3), (4) and $q_{i,\tau} \geq 0$ for all $\tau \in [t, T)$. The optimization problem for player j is symmetric to that of player i as $V^j(\tilde{y}_t, t) = V^i(-\tilde{y}_t, t)$. The corresponding Hamilton-Jacobi-Bellman (HJB) equation for player i is

$$0 = \max_{q_{i,t} \geq 0} \left[-\frac{c_i q_{i,t}^2}{2} + V_y^i \cdot (q_{i,t} - q_{j,t}) + V_t^i + \frac{V_{yy}^i}{2} \lambda \bar{S}^2 \right].$$

By definition, we have $\lambda \bar{S}^2 = \sigma^2$. Assuming an interior solution, substituting the first-order conditions $q_{i,t} = V_y^i / c_i$ and $q_{j,t} = -V_y^j / c_j$ yields the system

$$\begin{aligned} \frac{1}{2c_i}(V_y^i)^2 + \frac{1}{c_j} V_y^i V_y^j + V_t^i + V_{yy}^i \frac{\sigma^2}{2} = 0 \\ \frac{1}{2c_j}(V_y^j)^2 + \frac{1}{c_i} V_y^j V_y^i + V_t^j + V_{yy}^j \frac{\sigma^2}{2} = 0 \end{aligned}$$

subject to boundary conditions $V^i(-\infty, t) = 0$, $V^i(+\infty, t) = \theta$, $V^j(-\infty, t) = \theta$, $V^j(+\infty, t) = 0$, $V^i(\tilde{y}_T, T) = \theta \cdot 1_{\tilde{y}_T > 0}$ and $V^j(\tilde{y}_T, T) = \theta \cdot 1_{\tilde{y}_T < 0}$.

The Nash equilibrium is summarized in the following theorem. We include a simplified version of the proof in the appendix.

Theorem 1 (**Ryvkin 2022**). *In the Markov perfect equilibrium, the players' efforts in state $m_{i(j)}(\tilde{y}_t, t) : \mathbb{R} \times [0, T) \to \mathbb{R}_+$ are given by*

$$m_{i(j)}(\tilde{y}_t, t) = \frac{e^{-z^2/2}}{\sqrt{2\pi\sigma^2(T-t)}} \cdot \frac{\sigma^2}{2} \left[\gamma(\rho_i) + \gamma(\rho_j)\right] \left[1 - \rho(z)^2\right] \left[1 \pm \rho(z)\right] \tag{6}$$

where $z = \tilde{y}_t / (\sigma\sqrt{T-t})$, $\rho(z) = \gamma^{-1}\left(\Phi(z)\left[\gamma(\rho_i) + \gamma(\rho_j)\right] - \gamma(\rho_j)\right)$ and

$$\gamma(u) = \frac{u}{1-u^2} + \frac{1}{2} \ln \frac{1+u}{1-u}, \quad u \in (-1, 1) \tag{7}$$

$$\rho_i = \frac{e^{w_i} + e^{-w_j} - 2}{e^{w_i} - e^{-w_j}}, \quad \rho_j = \frac{e^{w_j} + e^{-w_i} - 2}{e^{w_j} - e^{-w_i}}, \quad w_{i(j)} = \frac{\theta}{\sigma^2 c_{i(j)}}.$$

Moreover, the total expected effort at time t, defined by $M(\tilde{y}_t, t) := \mathbb{E}\left[\int_t^T m_i(\tilde{y}_s, s) + m_j(\tilde{y}_s, s) ds\right]$, is given by $M(\tilde{y}_t, t) = \sigma\sqrt{T-t}\, q(z)$, where $q(z)$ is determined by the ordinary differential equation

$$\begin{aligned} q''(z) + \left[4A_+(z)e^{-z^2/2} + z\right] q'(z) - q(z) + 4A_-(z)e^{-z^2/2} = 0 \\ A_-(z) = \frac{\gamma(\rho_i) + \gamma(\rho_j)}{2\sqrt{2\pi}}[1 - \rho(z)^2], \quad A_+(z) = A_-(z)\rho(z) \end{aligned} \tag{8}$$

subject to the boundary conditions $q(z) \to 0$ as $z \to \pm\infty$. Specifically, when $c_i = c_j$, we have $w_i = w_j = w$ and the total expected effort is reduced to

$$M(\tilde{y}_t, t) = \frac{2w\sigma\sqrt{T-t}}{\sqrt{2\pi}} e^{-z^2/2} + o(w) \tag{9}$$

REMARK 1. The approximation is accurate when $w = \theta/(\sigma^2 c)$ is small, i.e., when uncertainty σ is high relative to the prize-cost ratio θ/c. This is the regime where the first-order expansion in w is most informative.

In equation (6), w_i and w_j summarize the players' abilities, and ρ_i and ρ_j are their normalized counterparts. The equilibrium effort $m_{i(j)}$ can be decomposed into two components: a normal density term $\phi(\tilde{y}_t; 0, \sigma^2(T-t))$ and an amplitude term $K_{i(j)}(z;\theta,\sigma,c_i,c_j) := \frac{\sigma^2}{2}\left[\gamma(\rho_i)+\gamma(\rho_j)\right]\left[1-\rho(z)^2\right]\left[1\pm\rho(z)\right]$. The density term governs how competitive intensity is distributed over the state space, whereas the amplitude term governs the asymmetry of effort across players.

This representation yields three observations useful for the empirical analysis. First, the parameters θ, c_i, and c_j enter the equilibrium effort expression through the amplitude term, so ability and prize incentives primarily affect the level and asymmetry of effort. Second, σ affects both the dispersion of the density term and the amplitude term, so its role is not reducible to a simple scale effect. Third, as the remaining time $T-t$ shrinks, effort becomes increasingly concentrated near the pivotal state.

The equilibrium characterization is inherited from Ryvkin (2022) after the filtered-state reduction to $(\tilde{y}_t, t)$ under $S_t \equiv \bar{S}$. Additional properties of the equilibrium are discussed in Ryvkin (2022). Here, we restrict attention to a brief design discussion under the maintained steady-state specification.

2.1. Implications on the Contest Design

In online crowdsourcing contests, designers typically control key decision variables such as the prize amount θ (we exclude more complex prize structures in this study), the contest duration T, and the signal precision λ. A key concern is that these variables may act as substitutes—implying that the optimal choice of one may alter or offset the optimal choices of the others.

Under the maintained steady-state filtering assumption $S_t = \bar{S}$, signal precision λ does not enter total expected effort $M(\tilde{y}_t, t)$. This is a property of the simplified specification rather than a general claim about contest design. Because uncertainty is fixed at $\bar{S}$, information improvements do not create an additional uncertainty-reduction channel in equilibrium effort. In richer specifications with time-varying filtering uncertainty, λ can affect effort dynamics quantitatively and potentially qualitatively.

Next, we turn to the other two design variables. When the two contestants are of similar strength, i.e., $c_i = c_j = c$, the total expected effort admits a particularly simple approximation, as shown in equation (9). Suppose the contest designer's objective takes the form $\beta^T M(\tilde{y}_0, 0)^\alpha - \theta$, where $0 < \alpha < 1$ and $0 < \beta \le 1$. The parameter $\beta \in (0,1]$ captures the time cost or discounting associated with longer contests. The parameter $\alpha \in (0,1)$ captures diminishing returns from aggregate contest effort to the designer's value. Although higher effort increases the chance of obtaining a better solution, the marginal value of additional effort is likely to decline once the contest has already attracted substantial search activity. Thus, M^α should be interpreted as a reduced-form mapping from total expected effort to the designer's effective value from the contest. This concavity also ensures that the designer's benefit from increasing the prize does not grow without bound relative to the linear prize cost.

Let $\bar{M}(\tilde{y}_0, 0)$ denote the leading-order approximation to total effort obtained from equation (9) under the small-w approximation, where $w = \theta/(\sigma^2 c)$ and $z = \tilde{y}_0/(\sigma\sqrt{T})$. Then the designer's objective based on $\bar{M}(\tilde{y}_0, 0)$ is tractable for optimization over (θ, T).

PROPOSITION 1. *Consider the symmetric benchmark with $c_i = c_j = c$, and fix the initial perceived gap $\tilde{y}_0$, the innovation risk σ, and the effort cost parameter c. Let $\bar{M}(\tilde{y}_0, 0)$ denote the leading-order approximation to total expected effort defined in equation (9), and let $\Pi(\theta, T, \lambda) = \beta^T \bar{M}(\tilde{y}_0, 0)^\alpha - \theta$, where $0 < \alpha < 1$ and $0 < \beta \le 1$. Then the following statements hold:*

(1) The approximated total effort $\bar{M}(\tilde{y}_0, 0)$ is increasing in the prize amount θ and increasing in the contest duration T.

(2) For any fixed $T > 0$, the designer's objective Π is single-peaked in the prize amount θ. For any fixed $\theta > 0$, if $0 < \beta < 1$, Π is single-peaked in the contest duration T; if $\beta = 1$, Π is increasing in T.

(3) If the designer jointly optimizes over (θ, T) on $\theta > 0$ and $T > 0$, then for $0 < \beta < 1$, Π admits a unique interior maximizer $(\theta^\star, T^\star)$. If $\beta = 1$, Π is increasing in T, so a finite optimizer exists only if an upper bound on contest duration is imposed.

Moreover, both $\bar{M}(\tilde{y}_0, 0)$ and $\Pi(\theta, T, \lambda)$ are neutral with respect to the signal precision λ.

3. Estimation Framework

This section presents the data-generating process and the Bayesian estimation procedure. We first connect the theoretical model to the observable contest data and then describe the likelihood construction and prior specification.

3.1. Data-Generating Process

For each Kaggle contest, the observable data from Meta-Kaggle can be grouped into three components.

The first component contains contest-level institutional features: contest duration, prize structure, and information disclosure rules. To match the two-player winner-take-all structure of the model, we restrict the empirical application to contests where a dominant top rivalry is plausible and monetary prize incentives can be mapped to an effective top-two winner-gain structure.

The second component records submission events of player i, denoted by $\{\hat{t}^i_k\}_{k=1}^{N_i}$, where N_i is the total number of submissions. Conditional on latent effort paths, we model submissions of players i and j as two conditionally independent inhomogeneous Poisson processes with intensities $\tau_i(t)$ and $\tau_j(t)$.[7] For any interval $\mathcal{S} \subset \mathcal{T}$, the expected number of submissions by player i is $\int_{\mathcal{S}} \tau_i(s)ds$. This proportionality condition should be interpreted as a measurement bridge from latent equilibrium effort to observed actions, rather than as a theorem of the game. For tractability, we assume

$$\tau_i(t) = r \cdot m_i(\tilde{y}_t, t) \tag{10}$$

where $r > 0$ scales effort into observed submission intensity.

The third component contains *public* and *private* leaderboard series, denoted by $\hat{x}^{i(j)}_t$ and $x^{i(j)}_t$. In Kaggle contests, public scores are computed from a released subset, while private scores are computed from withheld data and revealed only after the contest ends. We define the latent gap as $y_t := x^i_t - x^j_t$ and the observed public gap as $\hat{y}_t := \hat{x}^i_t - \hat{x}^j_t$. The public gap is the empirical counterpart of the signal process in (2):

$$dZ_t = \hat{y}_t dt \tag{11}$$

Combining (2) with (1) yields a heuristic observation equation for the public gap. The signal process induces a heuristic white-noise observation term, which we write as

$$\hat{y}_t = \int_0^t \left[m_i(\tilde{y}_s, s) - m_j(\tilde{y}_s, s)\right] ds + \sigma W_t + \frac{\xi_t}{\sqrt{\lambda}} \tag{12}$$

where ξ_t denotes the heuristic white-noise term induced by the signal process, σW_t captures accumulated innovation noise and λ controls signal precision. Equation (12) should be interpreted as continuous-time motivation for the discretely observed public-gap process used in estimation. At observation frequency Δ, the white-noise notation ξ_t is heuristic, with $\xi_t \approx (B_{t+\Delta} - B_t)/\Delta$ in implementation. Accordingly, the likelihood in Section 3 is built from discretely sampled observations rather than from a literally differentiable data path.

[7] That is, conditional on $\tau_i(t)$ and $\tau_j(t)$, the point processes $\{\hat{t}^i_k\}_{k=1}^{N_i}$ and $\{\hat{t}^j_k\}_{k=1}^{N_j}$ are independent.

The latent state, observed signal, and perceived state are jointly determined. By (3) and (11), players' perceived gap evolves as

$$d\tilde{y}_t = [m_i(\tilde{y}_t, t) - m_j(\tilde{y}_t, t)]\, dt + \sqrt{\lambda}\sigma(\hat{y}_t - \tilde{y}_t)dt \tag{13}$$

with initial condition $\tilde{y}_0 = \mu_0$. Equation (13) highlights the core behavioral channel: beliefs drift with expected relative effort and are corrected toward the observed public leaderboard at a speed governed by signal informativeness.

We denote the set of unknown parameters by $\Theta := \{c_i, c_j, \sigma, \lambda, \mu_0\}$. Here, μ_0 primarily captures initial latent alignment and is treated as a nuisance initial-condition parameter rather than a structural design primitive. Once Θ is specified, equations (1), (12), and (13) define the data-generating system, while realized trajectories remain stochastic.

3.2. Modeling Considerations

This subsection discusses three practical modeling concerns that are first-order for interpretation: strategic submission timing, the two-player approximation, and the normality assumption. The goal is to clarify the scope of the baseline framework and connect each concern to the robustness analyses that follow.

Strategic submission timing. Beyond the uncertainty caused by partial data disclosure, an additional source of signal noise stems from the timing of leaderboard updates: rankings are only refreshed upon new submissions, rendering the leaderboard uninformative during periods of inactivity. This lag can bias inference if participants deliberately time or withhold submissions.

Our baseline observation equation assumes submissions are driven by latent effort and does not explicitly model strategic withholding. Strategic submission timing is a first-order concern in the literature. For example, Lemus and Marshall (2021) adopt a similar simplification by excluding such behavior from their model. Our estimates should therefore be interpreted for contests where active updating behavior dominates withholding behavior. As a partial mitigation, we require a minimum number of submissions when selecting representative players (Section 5.1), which reduces cases where sparse strategic timing dominates observed dynamics.

Selection of Two Players. Another concern is the two-player approximation. Real contests involve many participants, so reducing the environment to two players changes strategic externalities and information structure. Our interpretation is that the model isolates the dominant top rivalry in contests where competition is concentrated near the frontier. In Section 6.1, we conduct a series of robustness checks. By selecting the top two competitors using alternative criteria, we aim to assess whether the earlier conclusions remain valid.

Normality. Another first-order concern is the normality assumption. Specifically, we rely on normal distributions in two critical components: the signal noise and the innovation shock. As discussed in Section 5.1, this assumption depends on transforming raw leaderboard percentages into the state variable $\hat{y}_t$. Under a uniform transformation, roughly half of contests do not conform well to the normality assumption, so this is not a routine technical caveat. In Section 6.2, we discuss this caveat and calibrate interpretation of the validation results.

3.3. Bayesian Estimation Framework

Our objective is to develop a Bayesian framework for inferring the unknown parameter set $\Theta := \{c_i, c_j, \sigma, \lambda, \mu_0\}$ using the data introduced above.

The key design principle is informational consistency: estimation uses only contest-phase information available to participants (submission timing and public leaderboard updates), while private leaderboard data are withheld for validation.

We discretize time into uniform intervals of length Δ. By definition, the likelihood function corresponding to the submission times of player i, $\{t_k^i\}_{k=1}^{N_i}$, is given as follows (a symmetric formulation applies to player j):

$$p\left(\{\hat{t}_k^i\}_{k=1}^{N_i} | \tau_i, \Theta\right) = \exp\left\{-\int_{s\in\mathcal{T}} \tau_i(s)ds\right\} \prod_{k=1}^{N_i} \tau_i(\hat{t}_k^i) \tag{14}$$

where τ_i is defined in (10) and r is specified exogenously. Under discretization, the integral term is evaluated numerically.

Next, suppose the public leaderboard gap $\hat{y}_t$ is sampled at times $(t_1, t_2, ..., t_N)$, yielding observations $\{\hat{y}_{t_k}\}_{k=1}^N$. Let $t_0 = 0$ and suppose the initial gap satisfies $y_0 = 0$. Then, according to the assumed data-generating process of $\hat{y}_t$ in (12), the corresponding likelihood function is

$$p\left(\{\hat{y}_{t_k}\}_{k=1}^N | \{\hat{t}_k^{i(j)}\}_{k=1}^{N_{i(j)}}, m_i, m_j, \Theta\right) = \phi\left(\{\hat{y}_{t_k}\}_{k=1}^N | \{\mu_k\}_{k=1}^N, \Sigma_y + \frac{I_N}{\Delta\lambda}\right) \tag{15}$$

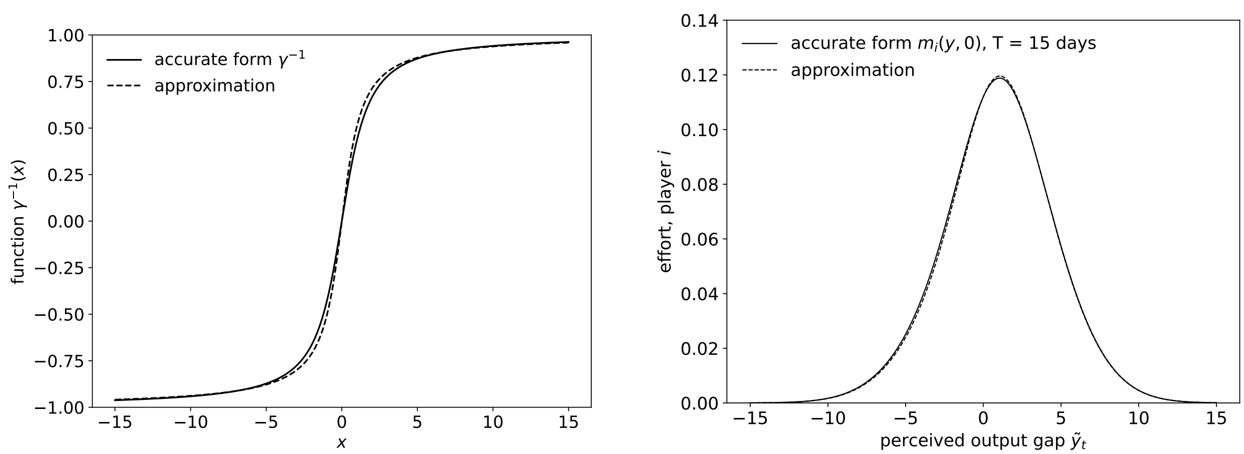
where $\mu_k = \int_0^{t_k} [m_i(\tilde{y}_s, s) - m_j(\tilde{y}_s, s)]\, ds$ and $\Sigma_y(i,j) = \sigma^2 \min(t_i, t_j)$. Moreover, let $N = N_i + N_j$, indicating that the public leaderboard is updated once a new submission occurs.

It is worth noting that, although the private leaderboard trajectory y_t is observable ex post in Meta-Kaggle, we deliberately exclude it from the estimation likelihood. In our framework, agents form beliefs about y_t based solely on $\hat{y}_t$, and their actions are driven by these beliefs. Conditioning on the realized values of y_t in estimation would effectively bypass the agent's informational constraints and undermine the role of λ in shaping belief formation. More importantly, it would destroy the out-of-sample validation design based on private leaderboard data. Treating y_t as latent preserves the model's behavioral interpretation and enables meaningful post-estimation validation.

To evaluate (14) and (15), we recursively construct $\tilde{y}_t$ and implied efforts (m_i, m_j) from (6) and (13). More precisely, conditional on a parameter vector and the observed public path, latent perceived states and implied effort paths are computed deterministically, while the data themselves remain stochastic. However, evaluating $m_{i(j)}(\tilde{y}_t, t)$ via equation (6) requires numerically approximating the inverse function γ^{-1}, which poses challenges for the use of gradient-based Markov Chain Monte Carlo (MCMC) sampling methods, such as Hamiltonian Monte Carlo (HMC, Neal 1996, Neal 2011, Betancourt 2017) and the No-U-Turn Sampler (NUTS, Hoffman and Gelman 2014). Hence, we approximate this inverse function with an analytical form:

$$\gamma^{-1}(x) \approx \frac{2}{\pi} \arctan(a \cdot x) \tag{16}$$

where a is around 0.947 by minimizing the 1-norm of the difference between the numerical inverse of $\gamma(\cdot)$ and the approximation $\frac{2}{\pi}\arctan(ax)$.[8] Figure 3 compares the equilibrium effort function $m_i(\tilde{y}_t, t)$ derived from the approximate analytical form of γ^{-1} in equation (16) with that obtained from the numerically accurate solution. As illustrated, the approximation closely replicates the true function; the approximation error appears negligible relative to posterior uncertainty over the calibration range considered.



(a) Auxiliary Function $\gamma^{-1}(x)$ (b) Equilibrium Effort $m_i(\tilde{y}_t, t)$

Figure 3 Comparison of the Accurate and Approximate Forms

Note: Panel (a) compares the exact inverse function $\gamma^{-1}(x)$ of the equilibrium component in (7) with its analytical approximation $\frac{2}{\pi}\arctan(ax)$ by (16), where $a \approx 0.947$ minimizes the 1-norm approximation error. Panel (b) compares the resulting equilibrium effort $m_i(\tilde{y}_t, t)$ at $t = 0$ (with $T = 15$, $\theta = 1$, $c_i = c_j = 1$, $\sigma = 1$) obtained from the analytical approximation versus the exact numerical solution.

We use truncated normal priors with explicit supports: $c_i, c_j \sim \mathcal{TN}(0.5, 5; [0.1, 5])$, $\sigma \sim \mathcal{TN}(1, 5; [0.5, 10])$, $\lambda \sim \mathcal{TN}(1, 5; [10^{-6}, 10])$, and $\mu_0 \sim \mathcal{TN}(\hat{y}_0, 1; [-20, 20])$. Thus, priors are weakly informative for $(c_i, c_j, \sigma, \lambda)$, while μ_0 is intentionally more informative to stabilize initialization.

The following lemma formalizes identification under the maintained specification. Identification comes from complementary moments: (c_i, c_j) shape asymmetric effort and submission timing

[8] Under infinity norm, the parameter $a = 0.856$; under 2-norm, the parameter $a = 0.895$.

through the point-process likelihood; σ loads on public-gap covariance over time; λ governs signal-noise intensity in public observations; and μ_0 anchors early-path perceived-state dynamics.

LEMMA 1. *The contest parameters c_i, c_j, σ, λ and μ_0 are jointly identifiable.*

4. Synthetic Experiments

Before applying our estimation procedure to real-world contest data, we evaluate its potential on synthetically generated data. These synthetic exercises serve three purposes: illustrating the data-generating mechanism, assessing finite-sample parameter recovery, and clarifying which parameters are easier or harder to recover under realistic data conditions.

Algorithm 1 Synthetic Data Simulation

```
Input: Δ, T, c_i, c_j, σ, λ, r, τ*, y_0, μ_0
Sample {s_k^{i(j)}}_{k=1}^{N_{i(j)}} from Poisson process (τ*) on [0, T)
Sample {u_k^{i(j)}}_{k=1}^{N_{i(j)}} from uniform distribution on [0, 1]
Sample series of Brownian motions (W_t) and (B_t)
Initialize ℓ^{i(j)} = 0, ŷ_0 = 0, ỹ_t = μ_0
for t = 0 to T do
    m_{i(j)}(ỹ_t, t) ← (6), τ^{i(j)}(t) ← (10)          {Use θ, c_{i(j)}, σ, r, T}
    y_{t+Δ} ← (1); ỹ_{t+Δ} ← (13)                      {Use σ, λ, Δ, W_{t+Δ}, y_0}
    for s_k^{i(j)} ∈ [t, t+Δ) do
        if u_k^{i(j)} < τ_{i(j)}(s_k^{i(j)})/τ*_{i(j)} then
            t̂_ℓ^{i(j)} ← s_k^{i(j)}; ℓ^{i(j)} ← ℓ^{i(j)} + 1   {Accept the submission event}
            ŷ_{t+Δ} ← (12)                             {Use σ, λ, B_{t+Δ}}
        end if
    end for
end for
Output: (y_t), (ỹ_t), (m_{i(j)}(ỹ_t, t)), {t̂_k^{i(j)}}_{k=1}^{N_{i(j)}}, (ŷ_{t_k})_{k=1}^{N_i+N_j}
```

A central challenge in generating synthetic data lies in dynamically constructing a point process that conforms to an inhomogeneous Poisson process. We first generate candidate submission events according to a homogeneous Poisson process over the whole contest duration, using a fixed high intensity $\tau^\star \geq \sup_t \tau_{i(j)}(t)$. Then, we apply the classical thinning procedure (Lewis and Shedler 1979) to determine whether each candidate event is accepted or not. In plain terms, Algorithm 1 generates candidate submissions, simulates latent and perceived states, accepts submissions via thinning, updates public leaderboard observations, and produces the synthetic sample used in the subsequent Bayesian recovery exercise.

As a baseline example, we consider a synthetic contest with a three-month horizon. The two participants differ slightly in effort costs, with $c_i = 1.2$ and $c_j = 1.5$. The remaining parameters are set to $\theta = 1$, $\sigma = 2.0$, $\lambda = 1.0$, and $r = 15$. Under this specification, we simulate the latent output gap, the public leaderboard path, the players' submission times, the perceived output gap, and the corresponding equilibrium effort paths.

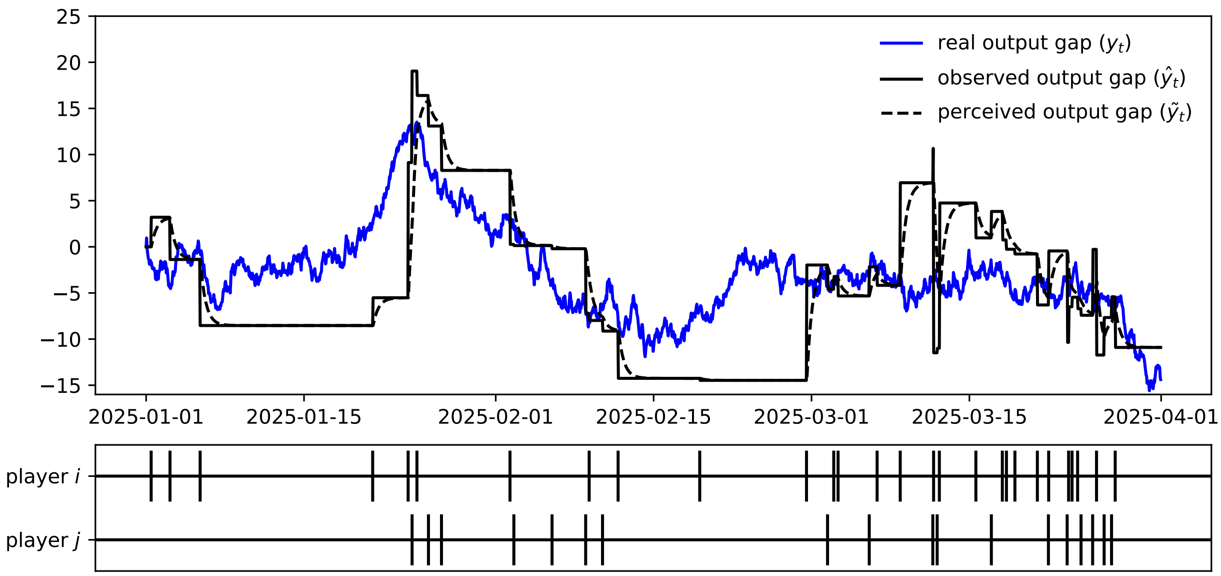


Figure 4 Synthetic Contest Data over a Three-Month Horizon

Note: The blue line represents the latent output gap y_t (private leaderboard), the solid black line represents the public leaderboard $\hat{y}_t$ (updated only upon submissions), and the dashed black line represents the perceived output gap $\tilde{y}_t$ inferred via Bayesian filtering. Vertical ticks mark submission times: ticks above the axis are player i's submissions; ticks below are player j's. Parameters: $\theta = 1.0$, $c_i = 1.2$, $c_j = 1.5$, $\sigma = 2.0$, $\lambda = 1.0$, $r = 15$, $T = 3$ months.

Figure 4 shows one realization of the baseline synthetic contest. The blue line represents the latent output gap y_t, which can be interpreted as the *private* leaderboard, while the solid black line represents the *public* leaderboard $\hat{y}_t$, updated whenever either player submits. The dashed black line represents the perceived output gap $\tilde{y}_t$, inferred from the public leaderboard as in equation (13). The short vertical ticks mark submission times generated by the inhomogeneous Poisson processes. In this realization, player i submits 28 times and player j submits 18 times. Between submissions, filtering uncertainty would generally evolve over time; for tractability, however, the synthetic design follows the maintained steady-state specification used throughout the model.

We use the same truncated-normal prior specification as in Section 3. The only point worth emphasizing here is that the prior on μ_0 is intentionally more informative, reflecting the weak finite-sample information about the initial latent state and helping stabilize initialization in the synthetic recovery exercises.

Table 1 presents the Bayesian inference results based on the synthetic data that we display in Figure 4. With limited data from a short contest, recovery is uneven across parameters. In this baseline case, c_i, c_j, and λ are reasonably recoverable, while σ is harder to recover and μ_0 remains weakly pinned down (posterior standard error close to 1). This finite-sample pattern is informative and motivates the richer-data exercises below.

Table 1 Bayesian Estimates from Synthetic Data

Parameters (true val.)	c_i 1.2	c_j 1.5	σ 2.0	λ 1.0	μ_0 0.0
Posterior Mean	1.089	1.751	2.801	1.105	0.005
Posterior Stderr	0.258	0.476	0.503	0.299	0.990
RMSE	0.281	0.538	0.945	0.317	0.990
95% Interval	[0.68, 1.68]	[1.01, 2.85]	[1.95, 3.93]	[0.62, 1.78]	[-1.92, 1.94]

Note: This table reports the posterior means, posterior standard errors, RMSE, and 95% credible intervals from Bayesian estimation using the synthetic contest data shown in Figure 4. In this single synthetic realization, players i and j submitted 28 and 18 times, respectively. RMSE is the square root of the MSE, where $\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = [\mathbb{E}(\hat{\theta}) - \theta]^2 + \text{Var}(\hat{\theta})$ by the bias–variance decomposition. The 95% credible interval reports the central posterior interval containing 95% of the posterior mass.

We next examine how recovery changes with richer data.

4.1. Parameter Recovery Under Richer Data

This subsection provides simulation-based recovery diagnostics under two richer-data regimes: pooling replicated contests under fixed structural parameters, and observing longer contests with more submissions. The goal is practical rather than asymptotic or theorem-based: to assess whether recovery improves in sensible ways as information accumulates.

Replications. To begin, we investigate recovery when multiple independent realizations of the contest are observed. Multiple contests are independently simulated under identical parameters, with their data pooled to form a larger sample for Bayesian inference. This allows us to assess whether Bayesian inference consistently recovers the true parameters as the number of observed contests increases.

The experimental results are presented in Table 2. We replicate the contest simulation with 10 and 20 independent instances. Compared with Table 1, RMSE and posterior uncertainty decline for c_i, c_j, σ, and λ, showing that the likelihood contains meaningful identifying content that compounds as more contests are added. In contrast, μ_0 remains weakly identified even with pooled contests.

Table 2 Bayesian Estimates from Pooled Synthetic Data

Parameters	c_i	c_j	σ	λ	μ_0
(true val.)	1.2	1.5	2.0	1.0	0.0
Pool of 10 Contests					
Posterior Mean	1.254	1.686	1.912	1.053	-0.009
Posterior Stderr	0.087	0.128	0.095	0.090	0.975
RMSE	0.102	0.225	0.130	0.104	0.975
Pool of 20 Contests					
Posterior Mean	1.204	1.639	1.981	0.988	0.008
Posterior Stderr	0.060	0.089	0.068	0.060	1.003
RMSE	0.060	0.165	0.071	0.062	1.003

Note: This table reports the posterior means, posterior standard errors, RMSE, and 95% credible intervals from Bayesian estimation using *pooled* data from multiple independent synthetic contest replications under identical structural parameters. For the pool of 10 contests, the total number of submissions is 280 by player i and 180 by player j; for the pool of 20 contests, the totals are 560 and 360, respectively. RMSE is the square root of the MSE, where $\mathrm{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta}-\theta)^2] = [\mathbb{E}(\hat{\theta})-\theta]^2 + \mathrm{Var}(\hat{\theta})$ by the bias–variance decomposition. The 95% credible interval reports the central posterior interval containing 95% of the posterior mass.

Table 3 Bayesian Estimates from Long Term Synthetic Data

Parameters	c_i	c_j	σ	λ	μ_0
Contest of 6 Months					
True Value	0.6	0.75	2.0	1.0	0.0
Posterior Mean	0.806	0.936	1.685	1.116	0.011
Posterior Stderr	0.112	0.142	0.148	0.171	0.986
RMSE	0.234	0.234	0.348	0.207	0.986
Contest of 12 Months					
True Value	0.3	0.375	2.0	1.0	0.0
Posterior Mean	0.333	0.436	2.067	1.101	0.012
Posterior Stderr	0.027	0.046	0.097	0.138	0.994
RMSE	0.043	0.076	0.118	0.171	0.994

Note: This table reports the posterior means, posterior standard errors, RMSE, and 95% credible intervals from Bayesian estimation using synthetic contest data with longer horizons (6 and 12 months). In the 6-month contest, players i and j submitted 59 and 51 times, respectively; in the 12-month contest, they submitted 100 and 97 times, respectively. To sustain active submission behavior over longer horizons, effort costs c_i and c_j are reduced proportionally (true values: $c_i = 0.6$, $c_j = 0.75$ for 6 months; $c_i = 0.3$, $c_j = 0.375$ for 12 months). RMSE is the square root of the MSE, where $\mathrm{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta}-\theta)^2] = [\mathbb{E}(\hat{\theta})-\theta]^2 + \mathrm{Var}(\hat{\theta})$ by the bias–variance decomposition. The 95% credible interval reports the central posterior interval containing 95% of the posterior mass.

Long-Horizon Contest. We then turn to a single-contest regime with a longer time horizon. Rather than increasing the number of trajectories, we examine how the accumulation of submissions over a longer period improves inference accuracy.

When generating long-horizon contest data, it is important to appropriately lower the effort costs c_i and c_j to ensure sustained effort from the contestants. This is because effort levels are influenced by the remaining time until the deadline: according to equation (6), the further away the deadline, the lower the equilibrium effort. Hence, this exercise is not a pure horizon comparison: we jointly vary horizon and effective submission intensity to keep the synthetic environment informative. Table 3 shows that posterior recovery improves as within-contest information increases, although this improvement reflects both a longer horizon and the lower effort costs used to maintain active submission behavior.

Taken together, the synthetic exercises indicate a consistent recoverability ranking: c_i, c_j, and λ are comparatively well recovered; σ is harder in thinner samples but improves with richer data; and μ_0 remains persistently weakly identified.

5. Empirical Application

In this section, we apply the model to real Kaggle contests and assess whether structurally recovered contest primitives align with private-leaderboard-based benchmarks.

We select representative Kaggle competitions from Meta-Kaggle and identify two focal participants i and j in each contest based on submission frequency, activity duration, and final ranking. Using their submission records and public leaderboard trajectories, we construct the observed data $\{t_k^i\}_{k=1}^{N_i}$, $\{t_k^j\}_{k=1}^{N_j}$ and $\{\hat{y}_{t_k}\}_{k=1}^{N_i+N_j}$ for contest-level Bayesian estimation under a unified prior specification.

The empirical design uses only contest-phase information observable to participants for Bayesian estimation, and then reserves *private* leaderboard information for validation. Because private leaderboard scores are generated by the same contest process but hidden during play, they provide a natural out-of-sample benchmark. Specifically, we compare structural contest-level estimates of λ and σ against benchmark quantities constructed from withheld private-leaderboard information. Figure 5 summarizes the data-model structure and validation design.

5.1. Preparing Data for Structural Estimation

To empirically implement the structural estimation framework developed in the previous sections using Kaggle data, the first step is to map the observed data onto the corresponding variables defined by the model. This subsection details the procedures for selecting competitions, identifying key participants, and incorporating contest-specific information such as prize structures to make the data compatible with the model's structural assumptions.

As a first selection criterion, we focus on monetary prize structure. We restrict attention to Kaggle competitions offering one, two, or three U.S.-dollar prizes. Competitions with a single prize naturally conform to the winner-take-all incentive structure. For competitions with two or three prizes, we align the empirical setting with the model by treating the prize difference between the top two participants as the effective gain from winning. Because non-winning participants may still receive symbolic recognition such as medals, focusing on the top-two prize gap helps

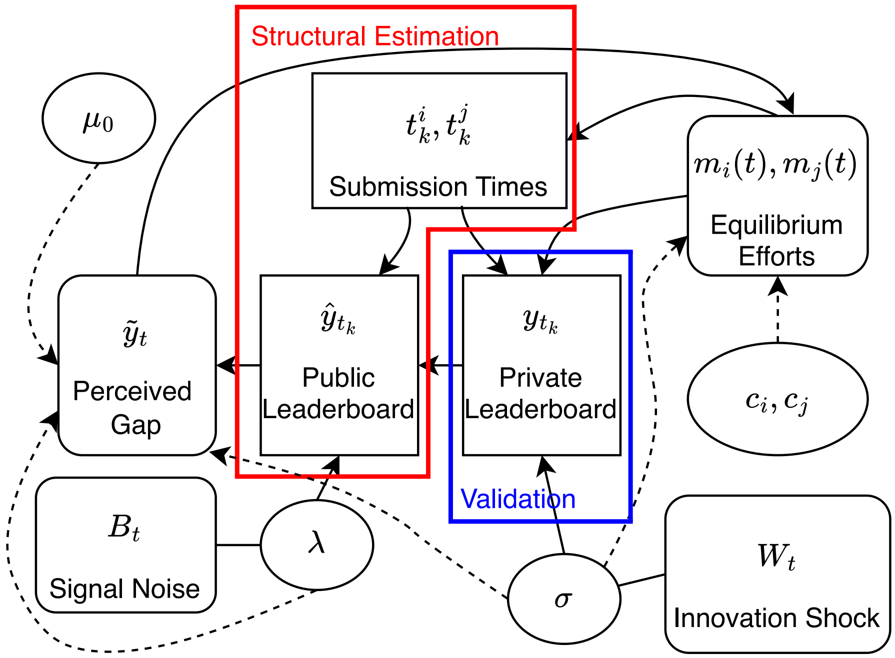


Figure 5 **Meta-Kaggle Dataset and the Model Structure**

Note: This figure shows the structure of a typical Kaggle contest and the information flow available to contestants. Rectangles represent observed data, rounded rectangles represent latent states, and ellipses represent contest parameters. Solid lines depict temporal relationships among time series variables and state variables, while dashed arrows show the influence of contest parameters on the data-generating process. The private leaderboard y_t is not observed by contestants during the contest and is excluded from estimation, serving only for out-of-sample validation.

mitigate confounding incentives and brings the empirical setting closer to the model's winner-take-all benchmark.

The second criterion requires that each contest allow for the identification of two top participants who engage in sustained and sufficiently intense competition throughout the contest period. We begin by examining the final rankings on the private leaderboard and consider candidates from the top downward. To ensure rich strategic interaction along with adequate data that supports reliable estimation, we impose two conditions: (1) each participant must have submitted at least five times, and (2) their active periods must exhibit sufficient temporal overlap. The first condition ensures that a participant appears frequently enough on the public leaderboard to attract the attention of potential rivals, while the second guarantees that the two players competed over a shared and extended timeframe. These criteria help ensure that the observed data are informative and that the underlying dynamics are well-suited for reliable structural Bayesian inference.

While analytically convenient, restricting attention to two participants may limit representativeness, oversimplify the strategic environment, and introduce selection concerns via *ex post* rankings. We therefore interpret this as a disciplined scope choice: a two-player structural approximation of the dominant rivalry near the top of contests with concentrated competition, frequent submissions, and overlapping activity windows.

The third criterion for contest selection concerns the format in which leaderboard scores are presented. Since Kaggle competitions are based on diverse real-world tasks, the scoring metrics vary accordingly. Some are based on raw scores (positive real numbers), while others are based on accuracy measures (expressed as percentages or decimals between 0 and 1). Ignoring these differences would lead to inconsistencies in units across contests, complicating any meaningful cross-contest comparison of parameter estimates.

Empirically, contests reporting accuracy-based scores (i.e., values between 0 and 1 on the leaderboard) tend to yield a higher success rate in identifying two focal players. Therefore, all contests used in this study are restricted to those where the leaderboard displays accuracy scores. To ensure consistency with the theoretical definition of the performance gap variable y_t and $\hat{y}_t$, we further transform the raw leaderboard accuracy values using the logit (log-odds) function:

$$y_t = a \cdot \log\left(\frac{p_t}{1-p_t}\right), \quad p_t \in [0.5, 1) \tag{17}$$

where p_t is the leaderboard score, and a is a scaling parameter in the transformation.[9] The logit function is S-shaped, with the key property that as the accuracy p_t approaches 1, each marginal improvement in p_t results in a disproportionately large increase in the transformed value y_t. This feature aligns with the intuition that it becomes increasingly difficult to improve performance near the upper bound, making such gains more informative in competitive settings.

5.2. Structural Validation

We select 73 contests from the Meta-Kaggle database, encompassing all eligible competitions concluded before April 15, 2025.[10] For each selected contest, we successfully identify two top participants who meet our criteria. Appendix C reports the Bayesian estimation results for each contest individually.

We next perform cross-validation of contest-specific primitives using private-leaderboard-based benchmarks.

Cross-Validation of Signal Precision λ. The signal precision parameter λ is contest-specific and depends on how informative public leaderboard feedback is relative to latent performance. We compare two estimates: (i) a structural estimate from submissions and public leaderboard paths, and (ii) a benchmark estimate from public-private leaderboard discrepancies. This benchmark is

[9] The scaling a can in principle be contest-specific to improve approximate normality after transformation. In this paper, to preserve comparability across contests and avoid extra tuning degrees of freedom, we fix $a = 1$ throughout.

[10] The Meta-Kaggle database is updated daily, with new competition data added after the contests conclude.

informative but not fully independent of the maintained structural setup, because it still relies on the transformed leaderboard scale used in the model.

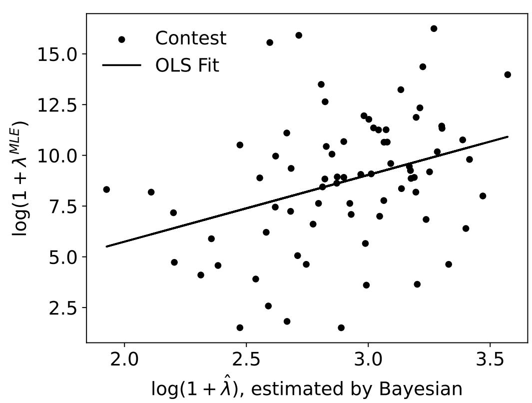
Let $\ell = 1, 2, \ldots$ index the contest. Under the maintained Gaussian discrepancy structure implied by (12), the public-private discrepancy yields the following benchmark maximum likelihood estimator (MLE) for λ_ℓ. Given that the discrepancy between the public and private leaderboard scores satisfies $\hat{y}_k^\ell - y_k^\ell \sim \mathcal{N}(0, 1/(\lambda_\ell \Delta))$, $\forall k = 1, 2, \ldots, N^\ell$, we have

$$\lambda_\ell^{\mathrm{MLE}} = \frac{N^\ell}{\Delta \sum_{k=1}^{N^\ell} (y_k^\ell - \hat{y}_k^\ell)^2}, \quad N^\ell = N_i^\ell + N_j^\ell$$

Here, N_i^ℓ and N_j^ℓ denote the numbers of submissions by players i and j in contest ℓ.

It is important to emphasize that this benchmark relies on the transformed-score normality approximation in (17). The transformation is adopted for tractability and does not guarantee strict normality. While formal tests indicate departures from exact normality in some contests, the validation results that follow are directionally robust to the choice of transformation parameter, suggesting that the benchmark remains informative despite this approximation.

We then regress the non-structural benchmark λ^{MLE} on the Bayesian structural estimate $\hat{\lambda}$ to examine whether a positive relationship holds between the two.



	coef.	stderr	95% CI
const.	-0.733	3.182	[-7.07, 5.61]
$\log(1+\hat{\lambda})$	3.264**	1.088	[1.10, 5.43]
Obs.		73	
R^2		0.110	
Adj. R^2		0.098	

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. (2) Normality diagnostics: Omnibus = 0.378, Jarque-Bera = 0.071, Skewness = -0.042, Kurtosis = 3.126. (3) Other diagnostic statistics: Condition Number = 28.2, Durbin-Watson = 2.244.

Figure 6: λ^{MLE} vs. Bayesian Estimate $\hat{\lambda}$

Note. This figure plots the benchmark maximum likelihood estimate λ^{MLE} against the Bayesian structural estimate $\hat{\lambda}$ for each of the 73 contests. λ^{MLE} is computed from public-private leaderboard discrepancies as $\lambda^{\mathrm{MLE}} = N/(\Delta \sum_k (y_k - \hat{y}_k)^2)$, where $N = N_i + N_j$ is the total number of submissions. The dashed line represents the fitted regression line from the model $\log(1+\lambda^{\mathrm{MLE}}) = \beta_0 + \beta_1 \log(1+\hat{\lambda}) + \varepsilon$. The positive slope ($\beta_1 = 3.264$, $p < 0.01$) indicates a statistically significant but modest association ($R^2 = 0.110$).

Figure 6 indicates a positive and statistically significant association between structural and benchmark estimates of λ. The slope on $\log(1+\hat{\lambda})$ is positive (3.264, $p < 0.01$), but fit is modest ($R^2 \approx 0.11$), so this should be interpreted as directional rather than high-precision contest-level recovery. This dispersion is consistent with limited sample size and with approximation error in benchmark construction, especially the normality-sensitive transformation step.

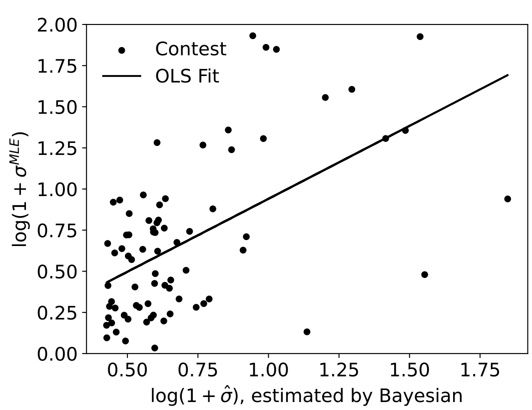
Cross-Validation of Innovation Risk σ. Another key contest-specific parameter is σ, which captures innovation uncertainty and the extent to which contest outcomes are driven by chance rather than effort. For each contest ℓ, we estimate $\hat{\sigma}_\ell$ structurally using the two players' public submission records $\{t_k^\ell\}_{k=1}^{N_i^\ell+N_j^\ell}$ and public leaderboard trajectories $\{\hat{y}_k^\ell\}_{k=1}^{N_i^\ell+N_j^\ell}$. We then construct a private-leaderboard-based benchmark for σ using score increments.

Under the diffusion specification in (1), the private-leaderboard increment satisfies $y_{k+1}^\ell - y_k^\ell \sim \mathcal{N}\left(\mu_k^\ell, \sigma^2(t_{k+1}^\ell - t_k^\ell)\right)$. Here, $\mu_k^\ell = \int_{t_k^\ell}^{t_{k+1}^\ell} \left[m_i^\ell(s) - m_j^\ell(s)\right] ds$. Given this, we derive the maximum likelihood estimator (MLE) of the contest-specific parameter σ (see Appendix B.2 for the derivation):

$$\sigma_\ell^{\mathrm{MLE}} = \left(\frac{1}{N_i^\ell + N_j^\ell} \sum_k \frac{\left[y_{k+1}^\ell - y_k^\ell - \hat{\mu}_k^\ell\right]^2}{t_{k+1}^\ell - t_k^\ell} \right)^{1/2}$$

where $\hat{\mu}_k^\ell$ is retrieved from Bayesian posterior effort trajectories. Hence, this benchmark is not fully model-free: it is less demanding on the signal side but still uses posterior drift adjustment. Therefore, this exercise should be interpreted as a consistency check using partial external information, not as a fully independent validation test.

As in the analysis above for λ, we regress σ^{MLE} on the structural estimate $\hat{\sigma}$ to examine whether a significant positive relationship exists between the two.



	coef.	stderr	95% CI
const.	0.054	0.116	[-0.18, 0.29]
$\log(1+\hat{\sigma})$	0.887***	0.152	[0.58, 1.19]
Obs.		73	
R^2		0.317	
Adj. R^2		0.308	

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. (2) Normality diagnostics: Omnibus = 1.280, Jarque-Bera = 0.774, Skewness = 0.225, Kurtosis = 3.214. (3) Other diagnostic statistics: Condition Number = 5.08, Durbin-Watson = 1.618.

Figure 7: σ^{MLE} vs. Bayesian Estimate $\hat{\sigma}$

Note. This figure plots the benchmark maximum likelihood estimate σ^{MLE} against the Bayesian structural estimate $\hat{\sigma}$ for each of the 73 contests. σ^{MLE} is computed from private leaderboard increments as $\sigma^{\mathrm{MLE}} = \left(\frac{1}{N}\sum_k \frac{(y_{k+1}-y_k-\hat{\mu}_k)^2}{t_{k+1}-t_k}\right)^{1/2}$, where $\hat{\mu}_k$ is retrieved from posterior effort trajectories. The dashed line represents the fitted regression line from the model $\log(1+\sigma^{\mathrm{MLE}}) = \beta_0 + \beta_1 \log(1+\hat{\sigma}) + \varepsilon$. The positive slope ($\beta_1 = 0.887$, $p < 0.001$) and $R^2 = 0.317$ indicate a stronger association than for λ, suggesting more reliable recovery of innovation risk.

Figure 7 reveals a stronger positive association than in the λ exercise. The intercept is small and insignificant, while the slope on $\log(1+\hat{\sigma})$ is 0.887 ($p < 0.001$), with $R^2 \approx 0.317$. Thus, innovation risk appears more reliably recovered across contests than signal precision in this sample. Taken together, the two exercises provide credible partial validation of key contest primitives, with materially stronger support for σ than for λ.

5.3. Counterfactual Policy Simulations

The validation exercise above suggests that the estimated primitives contain meaningful information about the contest environment. We use these primitives to revisit the design question in Section 2.1, where the designer's objective is motivated by $\beta^T M(\tilde{y}_0, 0)^\alpha - \theta$. In the empirical application, however, the designer's cardinal payoff is difficult to calibrate directly. We therefore use the model as a counterfactual diagnostic: rather than compare objective levels, we ask whether observed contest designs leave room for *prize-effort Pareto improvements*[11].

A counterfactual design generates a prize-effort Pareto improvement if, relative to the observed design, it weakly lowers prize spending and weakly increases predicted total effort, with at least one strict improvement. This is a comparison in the prize-effort outcome space, not a full welfare comparison across the organizer, participants, and consumers. For each contest, we hold fixed the estimated structural primitives and search for alternative prize-duration designs. To avoid extrapolating far beyond observed contest designs, we restrict the counterfactual search to a bounded redesign region, $\theta \in [0.01, \theta^{obs}]$ and $T \in [0.25T^{obs}, 3T^{obs}]$. The analysis is based on the 73 contests with complete observed design inputs and structural estimates.

Table 4 reports two complementary redesign exercises. The first asks how much predicted total effort can be increased without raising the prize above its observed level. By Proposition 1, this prize constraint binds, so the exercise fixes $\theta = \theta^{obs}$ and optimizes only over $T \in [0.25T^{obs}, 3T^{obs}]$. The average maximum effort gain is 96.31%, with a median of 81.53%.

The second exercise asks how much prize spending can be reduced while keeping predicted total effort weakly above the observed-design baseline. This cost-minimization problem jointly searches over θ and T. The average no-loss prize saving is 59.65%, with a median of 60.08%, corresponding to an average absolute reduction of 3.00 thousand USD.

These results suggest that, within the model and the bounded redesign region, many observed Kaggle designs are not located on the estimated prize-effort frontier. Adjusting prize size and contest duration could either raise predicted competitive effort, reduce prize expenditure, or both.

[11] We do not emphasize a baseline information-design counterfactual for λ. As shown in Section 2.1, under the maintained steady-state filtering restriction $S_t \equiv \bar{S}$, signal precision does not generate an independent comparative-static effect on total expected effort. Studying optimal information disclosure would require relaxing the steady-state restriction and allowing filtering uncertainty to evolve endogenously over time.

Because the effort gains in the first exercise may partly come from allowing duration to vary, we next conduct a prize-only counterfactual that holds T fixed. This provides a sharper test of whether observed prizes are excessive relative to the model-implied effort response.

Table 4 Counterfactual Effort Improvement and Prize Saving

Statistic	Effort improvement	Prize saving
Contests with positive margin	73/73	73/73
Mean percentage change	96.31%	59.65%
Median percentage change	81.53%	60.08%
95% CI for mean	$[85.46\%, 107.16\%]$	$[56.71\%, 62.60\%]$
One-sided t-statistic	17.70	40.37
One-sided t-test p-value	3.10×10^{-28}	1.69×10^{-51}
Wilcoxon signed-rank p-value	5.66×10^{-14}	5.66×10^{-14}
Minimum percentage change	32.05%	42.61%
Maximum percentage change	342.46%	91.18%
Mean absolute change	5.20	3.00
95% CI for absolute change	$[4.43, 5.97]$	$[2.17, 3.83]$

Note: This table reports counterfactual redesign results for the 73 contests with complete observed design inputs and structural estimates. For each contest, the observed design is (θ^{obs}, T^{obs}), and predicted total effort under this design serves as the baseline. The counterfactual search holds the estimated structural primitives fixed and jointly varies prize size and contest duration over $\theta \in [0.01, \theta^{obs}]$ and $T \in [0.25T^{obs}, 3T^{obs}]$. Prize is measured in thousand USD. The absolute effort change is in model-implied effort units; the absolute prize change is in thousand USD.

Effort Improvment: The maximum percentage increase in predicted total effort that can be achieved without increasing the prize above θ^{obs}.

Prize saving: The maximum percentage reduction in prize spending that can be achieved while keeping predicted total effort weakly above the observed-design baseline.

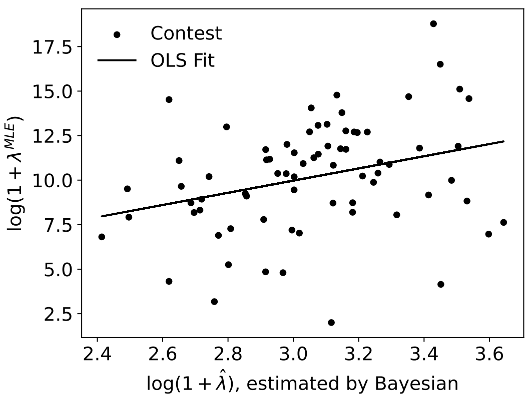
6. Robustness

This section examines the sensitivity of our main validation results to key modeling choices. We focus on two concerns highlighted in Section 3.2: the selection of the two-player focal pair and the normality assumption underlying the leaderboard transformation. For each robustness check, we re-estimate the model and recompute the benchmark validations, then compare the results to the baseline findings reported in Section 5.2.

6.1. Selection of Two Players

The baseline empirical analysis pairs the top two contestants in each contest. To assess whether the validation results are sensitive to this choice, we re-estimate the model under alternative focal-pair definitions.

As a robustness check, we replace the baseline pairing rule and estimate the model on the second- and third-ranked participants in each contest (based on final private leaderboard ranks). This specification shifts the focal rivalry away from the winner and tests whether the validation pattern is driven by one specific top-pair definition.



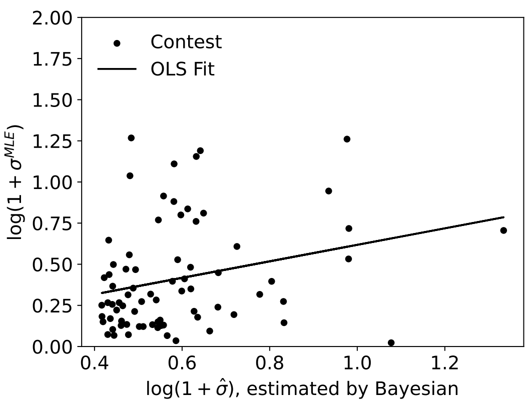
	coef.	stderr	95% CI
const.	0.290	3.819	[-7.91, 7.32]
$\log(1+\hat{\lambda})$	3.421***	1.244	[0.94, 5.90]
Obs.	73		
R^2	0.096		
Adj. R^2	0.084		

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. (2) Normality diagnostics: Omnibus = 3.192, Jarque-Bera = 2.424, Skewness = -0.411, Kurtosis = 3.350. (3) Other diagnostic statistics: Condition Number = 36.9, Durbin-Watson = 2.236.

Figure 8: Robustness Check (2nd–3rd Pair): λ^{MLE} vs. Bayesian Estimate $\hat{\lambda}$

Note. This figure replicates the validation exercise in Figure 6 using an alternative focal pair: the second- and third-ranked participants (based on final private leaderboard ranks) instead of the top two. The dashed line is the fitted regression $\log(1+\lambda^{\text{MLE}}) = \beta_0 + \beta_1 \log(1+\hat{\lambda}) + \varepsilon$, with $\beta_1 = 3.421$ ($p < 0.01$) and $R^2 = 0.096$. The positive slope remains significant, though the fit is modest, indicating directional robustness of the λ validation to the choice of focal pair.

For signal precision, the slope remains positive and significant, while fit remains modest ($R^2 \approx 0.10$). This mirrors the baseline pattern: directional robustness with limited contest-level precision.



	coef.	stderr	95% CI
const.	0.117	0.130	[-0.14, 0.38]
$\log(1+\hat{\sigma})$	0.501***	0.212	[0.08, 0.92]
Obs.	73		
R^2	0.071		
Adj. R^2	0.059		

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. (2) Normality diagnostics: Omnibus = 12.72, Jarque-Bera = 13.66, Skewness = 1.02, Kurtosis = 3.50. (3) Other diagnostic statistics: Condition Number = 7.84, Durbin-Watson = 1.87.

Figure 9: Robustness Check (2nd–3rd Pair): σ^{MLE} vs. Bayesian Estimate $\hat{\sigma}$

Note. This figure replicates the validation exercise in Figure 7 using an alternative focal pair: the second- and third-ranked participants (based on final private leaderboard ranks) instead of the top two. The dashed line is the fitted regression $\log(1+\sigma^{\text{MLE}}) = \beta_0 + \beta_1 \log(1+\hat{\sigma}) + \varepsilon$, with $\beta_1 = 0.501$ ($p < 0.05$) and $R^2 = 0.071$. The positive slope remains significant, though the association is weaker than in the baseline ($\beta_1 = 0.887$, $R^2 = 0.317$), indicating that σ recovery is directionally robust but sensitive to the choice of focal pair.

For innovation risk, the positive and significant association also survives this alternative pairing rule, although the fit is weaker than in the baseline specification. Together, these results suggest the main directional validation findings are not driven solely by one arbitrary top-two definition.

6.2. Normality Caveat and Interpretation

Section 5 already emphasizes that the logit transformation is adopted for tractability and does not guarantee strict normality for every contest. Empirically, the degree of approximate normality in the transformed data is affected by the choice of the logit scaling parameter a in equation (17). This issue is especially relevant for interpreting the λ benchmark, which relies more directly on Gaussian assumptions for public-private score discrepancies. In auxiliary checks not reported here, we find a qualitatively similar significance pattern across several reasonable choices of a. Additional results for alternative choices of a are available in our online GitHub code repository.

We do not conduct a fully non-Gaussian robustness analysis, because both the benchmark constructions and the Bayesian estimation procedure rely on the maintained Gaussian structure. Extending the analysis beyond this setting would require re-specifying the likelihood and re-deriving the benchmark estimators under alternative distributional assumptions. We leave this to future research.

Overall, the robustness analyses support a disciplined interpretation: the two-player reduction is a useful empirical summary of focal rivalry rather than a literal description of full contest dynamics, and model validation is strongest for innovation-risk recovery and more modest for signal-precision recovery.

7. Conclusion

This paper develops a dynamic game-theoretic model and a Bayesian structural estimation framework for online crowdsourcing contests under noisy public feedback. Using only contest-phase information observable to participants, we recover latent strategic objects and then validate contest-specific primitives with private-leaderboard-based benchmarks that are excluded from estimation. Empirically, across 75 Kaggle contests, the evidence supports credible partial validation, with materially stronger support for innovation risk σ than for signal precision λ. In complementary synthetic exercises, the initial-condition parameter μ_0 is comparatively weakly pinned down relative to the main contest-level primitives.

Nonetheless, several limitations deserve mention. First, the estimation relies on assumptions about the data-generating process (DGP) that may not fully reflect real-world complexities. For example: (1) many contests use multi-tier or team-based prizes rather than the winner-take-all format assumed here; (2) we focus on two-player settings, while actual contests involve broader participation; (3) leaderboard scores vary in format, complicating their integration into a unified signal structure. Second, the estimation method simplifies belief updating by assuming the inference variance quickly converges and remains constant. A more realistic approach would allow inference precision to evolve over time, e.g., via hybrid Kalman filtering, but such extensions would increase analytical complexity. These limitations point to valuable directions for future work, including incorporating richer prize structures, multi-agent dynamics, and more flexible learning mechanisms to enhance realism and applicability.

Appendix A: Proofs

Proof of Proposition 1 Under the symmetric benchmark $c_i = c_j = c$, equation (9) implies that the leading-order approximation to total expected effort at $t = 0$ is

$$\bar{M}(\tilde{y}_0, 0) = \frac{2\theta\sqrt{T}}{\sigma c\sqrt{2\pi}} \exp\left(-\frac{\tilde{y}_0^2}{2\sigma^2 T}\right).$$

For notational convenience, define $A = 2/(\sigma c\sqrt{2\pi})$ and $B = \tilde{y}_0^2/(2\sigma^2)$. Then

$$\bar{M}(\tilde{y}_0, 0) = A\theta T^{1/2} \exp(-B/T).$$

Since λ does not enter $\bar{M}(\tilde{y}_0, 0)$, the approximated total effort is neutral with respect to the signal precision λ. Moreover, because the designer's objective is given by $\Pi(\theta, T, \lambda) = \beta^T \bar{M}(\tilde{y}_0, 0)^\alpha - \theta$, it is also neutral with respect to λ. It remains to establish the dependence on θ and T.

We first prove the monotonicity of $\bar{M}(\tilde{y}_0, 0)$. Differentiating with respect to θ gives $\partial\bar{M}(\tilde{y}_0, 0)/\partial\theta = AT^{1/2}\exp(-B/T) > 0$, because $A > 0$ and $T > 0$. Hence, $\bar{M}(\tilde{y}_0, 0)$ is increasing in the prize amount θ. Next, differentiating with respect to T gives

$$\frac{\partial\bar{M}(\tilde{y}_0, 0)}{\partial T} = A\theta\exp(-B/T)\left(\frac{1}{2}T^{-1/2} + BT^{-3/2}\right).$$

Since $A > 0$, $\theta > 0$, $B \geq 0$, and $T > 0$, we have $\partial\bar{M}(\tilde{y}_0, 0)/\partial T > 0$. Therefore, $\bar{M}(\tilde{y}_0, 0)$ is increasing in the contest duration T.

We now turn to the contest designer's objective. Using the expression above, we can write

$$\Pi(\theta, T, \lambda) = A^\alpha\theta^\alpha\beta^T T^{\alpha/2}\exp(-\alpha B/T) - \theta.$$

First, fix $T > 0$. Define $D(T) = AT^{1/2}\exp(-B/T)$. Then $\bar{M}(\tilde{y}_0, 0) = D(T)\theta$, and the objective becomes

$$\Pi(\theta, T, \lambda) = \beta^T D(T)^\alpha\theta^\alpha - \theta.$$

Differentiating with respect to θ gives $\partial\Pi(\theta, T, \lambda)/\partial\theta = \alpha\beta^T D(T)^\alpha\theta^{\alpha-1} - 1$. The second derivative is

$$\frac{\partial^2\Pi(\theta, T, \lambda)}{\partial\theta^2} = \alpha(\alpha - 1)\beta^T D(T)^\alpha\theta^{\alpha-2} < 0,$$

because $0 < \alpha < 1$. Hence, for any fixed $T > 0$, the objective is strictly concave in θ. Moreover, as $\theta \downarrow 0$, we have $\theta^{\alpha-1} \to +\infty$, so $\partial\Pi/\partial\theta \to +\infty$; as $\theta \to +\infty$, we have $\theta^{\alpha-1} \to 0$, so $\partial\Pi/\partial\theta \to -1$. Therefore, there exists a unique maximizer in θ, and the objective is single-peaked in the prize amount θ.

Next, fix $\theta > 0$ and consider the dependence on T. Up to the constant term $-\theta$, the objective can be written as

$$\Pi(\theta, T, \lambda) + \theta = (A\theta)^\alpha\beta^T T^{\alpha/2}\exp(-\alpha B/T).$$

Let $G(T) = \log(\Pi(\theta, T, \lambda) + \theta)$. Since $\Pi(\theta, T, \lambda) + \theta > 0$, the sign of $\partial\Pi/\partial T$ is the same as the sign of $G'(T)$. We have

$$G(T) = \alpha\log(A\theta) + T\log\beta + (\alpha/2)\log T - \alpha B/T.$$

Therefore, $G'(T) = \log\beta + \alpha/(2T) + \alpha B/T^2$.

If $\beta = 1$, then $\log\beta = 0$, and hence $G'(T) = \alpha/(2T) + \alpha B/T^2 > 0$. Thus, when $\beta = 1$, the objective is increasing in T.

If $0 < \beta < 1$, then $\log\beta < 0$. Moreover,

$$G''(T) = -\alpha/(2T^2) - 2\alpha B/T^3 < 0.$$

Thus, $G'(T)$ is strictly decreasing in T. As $T \downarrow 0$, $G'(T) \to +\infty$, while as $T \to +\infty$, $G'(T) \to \log\beta < 0$. Hence, there exists a unique $T^\star > 0$ such that $G'(T^\star) = 0$. Since the sign of $\partial\Pi/\partial T$ is the same as the sign of $G'(T)$, the objective increases for $T < T^\star$ and decreases for $T > T^\star$. Therefore, if $0 < \beta < 1$, the objective is single-peaked in the contest duration T.

Finally, consider the joint optimization over (θ, T). For each fixed $T > 0$, the first-order condition with respect to θ is

$$\alpha A^\alpha \beta^T T^{\alpha/2} \exp(-\alpha B/T)\theta^{\alpha-1} = 1.$$

Because $0 < \alpha < 1$, this condition has a unique solution, denoted by $\theta^\star(T)$, given by

$$\theta^\star(T) = \left[\alpha A^\alpha \beta^T T^{\alpha/2} \exp(-\alpha B/T)\right]^{1/(1-\alpha)}.$$

The strict concavity in θ established above implies that $\theta^\star(T)$ is the unique optimal prize conditional on T.

We can therefore study the profile objective $\Pi(\theta^\star(T), T, \lambda)$. By the first-order condition, $A^\alpha \beta^T T^{\alpha/2} \exp(-\alpha B/T)(\theta^\star(T))^\alpha = \theta^\star(T)/\alpha$. Hence,

$$\Pi(\theta^\star(T), T, \lambda) = \frac{1-\alpha}{\alpha}\theta^\star(T).$$

Since $(1-\alpha)/\alpha > 0$, maximizing the profile objective over T is equivalent to maximizing $\theta^\star(T)$. Because the exponent $1/(1-\alpha)$ is positive, this is equivalent to maximizing $\beta^T T^{\alpha/2} \exp(-\alpha B/T)$ over $T > 0$.

Let $H(T) = T\log\beta + (\alpha/2)\log T - \alpha B/T$. Then maximizing $\beta^T T^{\alpha/2} \exp(-\alpha B/T)$ is equivalent to maximizing $H(T)$. Differentiating gives $H'(T) = \log\beta + \alpha/(2T) + \alpha B/T^2$, and

$$H''(T) = -\alpha/(2T^2) - 2\alpha B/T^3 < 0.$$

Thus, $H(T)$ is strictly concave on $T > 0$.

If $0 < \beta < 1$, then $\log\beta < 0$. Moreover, as $T \downarrow 0$, $H'(T) \to +\infty$, while as $T \to +\infty$, $H'(T) \to \log\beta < 0$. Therefore, there exists a unique $T^\star > 0$ such that $H'(T^\star) = 0$. The corresponding optimal prize is $\theta^\star = \theta^\star(T^\star)$. Hence, when $0 < \beta < 1$, the joint problem admits a unique interior maximizer $(\theta^\star, T^\star)$.

If $\beta = 1$, then $H'(T) = \alpha/(2T) + \alpha B/T^2 > 0$ for all $T > 0$. Therefore, the profile objective is increasing in T, and no finite interior maximizer in T exists unless an upper bound on contest duration is imposed. If an upper bound $T \le \bar{T}$ is imposed, the optimal duration is $T^\star = \bar{T}$, and the optimal prize is $\theta^\star(\bar{T})$. □

Proof of Lemma 1 This proof is a sketch under regularity conditions: bounded parameter space, positive-definite covariance matrices, and strictly positive inhomogeneous Poisson intensities on $\mathcal{T}$. Suppose two parameter vectors $\Theta = (c_i, c_j, \mu_0, \sigma, \lambda)$ and $\Theta' = (c_i', c_j', \mu_0', \sigma', \lambda')$ induce the same likelihood for all admissible observations.

First, consider the Gaussian leaderboard component in (15). Equality of Gaussian likelihoods for all sampled paths implies equality of both covariance and mean structures. Hence,

$$\Sigma_y + I_N/(\Delta\lambda) = \Sigma'_y + I_N/(\Delta\lambda') \quad \text{and} \quad \mu_k = \mu'_k \ \forall k.$$

Since $\Sigma_y(i,j) = \sigma^2 \min(t_i, t_j)$, covariance equality implies $\sigma = \sigma'$ and $\lambda = \lambda'$.

Second, consider the point-process component in (14). For inhomogeneous Poisson processes, equality of likelihoods for all event realizations implies equality of intensity functions almost everywhere, so

$$\tau_i(t) = \tau'_i(t), \qquad \tau_j(t) = \tau'_j(t) \quad \text{a.e. on } \mathcal{T}.$$

Because $\tau_{i(j)}(t) = r\, m_{i(j)}(\tilde{y}_t, t)$ and r is fixed, intensity equality implies equality of equilibrium effort paths.

With σ and λ already identified from the Gaussian part, equality of effort paths implies

$$q_{i,t}(\tilde{y}_t, t; c_i, c_j, \sigma, \lambda) = q'_{i,t}(\tilde{y}_t, t; c'_i, c'_j, \sigma, \lambda) \quad \text{a.e. on } \mathcal{T},$$

and analogously for player j. By the closed-form policy in (6), these equalities imply $c_i = c'_i$ and $c_j = c'_j$. In particular, once (σ, λ) are fixed, the effort policy coefficients are one-to-one in (c_i, c_j) through the mappings $w_{i(j)} = \theta/(\sigma^2 c_{i(j)})$ and $\rho_{i(j)} = \rho(w_{i(j)})$. Hence equality of effort paths over $\mathcal{T}$ pins down both cost parameters rather than only their ratio. Because the equilibrium effort functions load asymmetrically on c_i and c_j through w_i and w_j, equality of both players' effort paths rules out observational equivalence based only on a common scale change. Given $(c_i, c_j, \sigma, \lambda)$, the filtered dynamics in (13) are recursively determined conditional on the observed signal path, so equality of implied filtered paths for all admissible observations yields $\mu_0 = \mu'_0$. Therefore, $\Theta = \Theta'$, establishing joint identification. $\square$

Appendix B: Auxiliary Results

B.1. Solve S_t in Equation (4)

If S_t is in steady state, $dS_t/dt = 0 \Leftrightarrow S_t = \bar{S} \equiv \sigma/\sqrt{\lambda}$. If S_t is not in steady state, i.e., $S_t \neq \bar{S}$, we first isolate the two variables and get

$$\frac{dS}{\sigma - \lambda S^2} = dt$$

Then, we take the integral on both sides

$$t = \int \frac{dS}{\sigma - \lambda S^2} = \frac{1}{\sigma\sqrt{\lambda}} \int \frac{dS\sqrt{\lambda}/\sigma}{1 - (S\sqrt{\lambda}/\sigma)^2} \equiv \frac{1}{\sigma\sqrt{\lambda}} \int \frac{du}{1-u^2}$$

where $u = S\sqrt{\lambda}/\sigma = S/\bar{S}$. For $\tanh^{-1}(u)$ we require $|u| < 1$ ($S < \bar{S}$), and for $\coth^{-1}(u)$ we require $|u| > 1$ ($S > \bar{S}$). Hence,

$$\sigma\sqrt{\lambda} \cdot t = \begin{cases} \tanh^{-1}(u) - K_1, & \text{if } |u| < 1 \\ \coth^{-1}(u) - K_2, & \text{if } |u| > 1 \end{cases} = \begin{cases} \tanh^{-1}(S/\bar{S}) - K_1, & \text{if } S < \bar{S} \\ \coth^{-1}(S/\bar{S}) - K_2, & \text{if } S > \bar{S} \end{cases}$$

Thus, in the non-steady-state case,

$$S = \begin{cases} \bar{S} \cdot \tanh(\sigma\sqrt{\lambda} \cdot t + K_1), & \text{if } S < \bar{S} \\ \bar{S} \cdot \coth(\sigma\sqrt{\lambda} \cdot t + K_2), & \text{if } S > \bar{S} \end{cases}$$

Finally, we determine the constants K_1, K_2 by the initial condition S_0 and have

$$K_1 = \tanh^{-1}(S_0/\bar{S}), \quad K_2 = \coth^{-1}(S_0/\bar{S})$$

If $\lambda = 0$, we have $S_t = S_0 + \sigma^2 t$, i.e., the estimation variance is increasing in time linearly. If $\lambda > 0$, the solution of (4) is

$$S_t = \begin{cases} \bar{S} \cdot \tanh\left\{t \cdot \sigma\sqrt{\lambda} + \tanh^{-1}\left(S_0/\bar{S}\right)\right\} & \text{if } S_0 < \bar{S} \\ \bar{S} & \text{if } S_0 = \bar{S} \\ \bar{S} \cdot \coth\left\{t \cdot \sigma\sqrt{\lambda} + \coth^{-1}\left(S_0/\bar{S}\right)\right\} & \text{if } S_0 > \bar{S} \end{cases}$$

Specifically, $\bar{S} = \sigma/\sqrt{\lambda}$ when $\lambda > 0$ and $\bar{S} = \infty$ when $\lambda = 0$.

B.2. MLE for σ^2

Suppose we observe approximately independent data points $X_1, ..., X_n$, where each $X_k \sim \mathcal{N}(M_k/r, \sigma^2 T_k)$ with known constants M_k and T_k. The goal is to estimate the auxiliary parameters r and σ via maximum likelihood estimation (MLE). Ignoring constant terms, the log-likelihood function of the observed data is

$$\ell(r, \sigma^2) = -\frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{k=1}^{n}\frac{(X_k - M_k/r)^2}{T_k}.$$

To estimate r, we expand the summation of second term and have

$$Q(r) = \sum_{k=1}^{n}\left(\frac{X_k^2}{T_k} - \frac{2X_k M_k}{rT_k} + \frac{M_k^2}{r^2 T_k}\right) = A - \frac{2B}{r} + \frac{C}{r^2},$$

where $A = \sum X_k^2/T_k$, $B = \sum X_k M_k/T_k$, and $C = \sum M_k^2/T_k$. Taking the derivative with respect to r and setting it to zero gives:

$$\frac{dQ}{dr} = \frac{2B}{r^2} - \frac{2C}{r^3} = 0 \quad \Rightarrow \quad \hat{r} = \frac{C}{B} = \frac{\sum M_k^2/T_k}{\sum X_k M_k/T_k}.$$

This expression is well-defined when $B \neq 0$; under the model and positive T_k, the economically relevant case is $B > 0$, which yields $\hat{r} > 0$. Then, substituting $\hat{r}$ back into the likelihood, the MLE for σ^2 is obtained by maximizing

$$\ell(\sigma^2) = -\frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{k=1}^{n}\frac{(X_k - M_k/\hat{r})^2}{T_k},$$

which yields

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{k=1}^{n}\frac{(X_k - M_k/\hat{r})^2}{T_k}. \tag{18}$$

In the empirical application, this serves as an auxiliary benchmark approximation; dependence across leaderboard increments may weaken its exact finite-sample optimality.

Appendix C: Statistics

Table 5 reports raw contest-level estimates for transparency, even though some benchmark values are highly dispersed. Large values of λ^{MLE} can arise mechanically when public-private score discrepancies are very small, because the estimator is inverse-quadratic in those discrepancies. Accordingly, cross-contest comparisons of λ^{MLE} are most interpretable on a transformed scale (as used in the main-text regressions).

Table 5: Contest-Level Structural Estimates and Benchmarks

Contest Id	N_i	N_j	Prize (k, USD)	Public Data %	Bayesian Posterior Mean λ	σ	μ_0	c_i	c_j	r	Cross Validation λ^{MLE}	σ^{MLE}
2445	7	19	5.00	25	15.377	0.980	-0.089	4.304	3.154	2.088	1539.379	0.394
2454	19	13	0.15	62	15.016	1.030	4.377	0.688	0.810	18.035	932.836	0.659
2464	11	12	0.95	20	9.844	2.325	-0.474	1.631	1.380	14.217	99.556	3.741
2478	13	11	0.95	30	13.636	0.673	-0.444	3.171	3.811	6.333	19406.675	0.770
2489	12	21	0.50	10	16.650	0.576	0.431	3.535	2.303	11.210	5759.075	0.843
2549	13	15	0.50	30	12.214	1.357	-1.055	1.409	1.157	14.758	525.201	2.891
2551	48	30	1.50	30	26.143	0.539	-0.286	3.207	4.141	6.669	140015.172	0.512
2667	55	67	30.00	50	16.977	1.695	0.758	3.919	3.803	1.184	1.805	5.430
2749	22	22	3.00	42	20.708	0.701	0.906	4.308	4.022	4.367	37219.066	0.340
2762	12	21	1.00	35	11.659	0.779	1.068	3.103	2.237	13.005	20.923	1.244
2860	9	6	1.00	30	12.735	0.816	0.133	3.096	3.728	6.933	33538.355	0.034
3064	13	5	0.20	25	14.106	0.630	-0.788	1.160	2.780	13.821	8206710.757	0.263
3080	25	27	0.09	25	23.442	0.531	-1.217	0.623	0.643	19.327	150384.896	0.187
3288	20	10	1.00	0	10.864	3.729	-1.169	0.822	1.750	15.448	2.179	0.616
3338	31	29	1.50	30	24.827	0.541	0.203	3.508	3.345	7.687	8200.515	0.244
3353	26	17	10.00	30	16.307	1.232	-0.249	3.749	4.305	2.066	22001.196	1.410
3366	16	11	0.50	50	15.653	0.693	-1.936	2.055	3.018	13.340	4849.451	0.499
3507	9	13	0.50	33	14.030	1.055	0.529	2.440	1.891	14.095	163.652	1.101
3509	8	13	0.20	30	12.714	0.741	-0.024	2.491	1.513	13.807	3167.382	0.884
3526	15	13	5.00	20	15.809	0.793	-0.053	4.127	4.159	1.280	11651.555	0.242
3641	7	41	1.50	43	20.406	0.579	0.615	4.292	1.875	3.434	2447.287	0.319
3774	29	26	0.50	50	23.554	0.537	-0.997	2.580	2.898	13.300	29.891	0.952
3800	21	28	3.00	15	20.046	0.820	3.384	4.322	3.599	6.632	1174.496	0.625
3926	25	20	0.30	25	22.016	0.547	-0.047	1.237	1.529	13.393	4798.921	0.332
3928	15	21	0.68	70	19.525	0.584	-0.421	3.604	3.051	11.838	87393.056	0.139
3929	35	31	8.00	50	5.865	3.653	0.062	3.165	3.520	10.320	4794.303	5.862
3960	47	50	8.00	40	28.973	0.883	-0.992	4.624	4.670	3.583	624.827	0.514
4104	8	6	20.00	20	10.864	2.116	0.036	3.729	3.994	0.742	42409.966	0.141
4366	31	31	8.00	2	23.414	0.913	-0.434	4.367	4.554	2.979	3601.181	0.487
4407	44	51	4.00	19	14.573	1.670	0.070	3.591	3.238	12.165	56.026	2.693
4453	15	18	2.00	30	17.165	0.617	1.204	4.110	3.071	2.607	47662.222	0.892
4477	7	11	2.00	50	11.870	1.104	-3.305	3.867	3.222	5.142	2524.322	0.325
4488	11	8	2.00	1	8.033	5.347	1.108	2.160	2.843	12.739	1903.540	1.559
4493	19	14	10.00	40	16.683	1.161	-0.191	3.883	4.306	1.567	7787.118	0.355
4657	66	62	4.00	30	34.581	0.559	0.330	4.657	4.691	3.410	1221277.563	0.205
4699	13	22	5.00	30	18.734	0.765	0.112	4.242	3.833	1.207	223019.351	0.211
4986	16	5	10.00	50	13.378	1.203	-0.029	3.287	4.306	1.214	80756.892	0.393
5056	18	35	5.00	33	21.957	0.720	-0.780	4.484	4.094	2.763	536274.232	0.324
5144	11	9	20.00	20	7.244	3.118	10.283	3.617	3.737	1.411	3752.980	2.695
5174	42	47	3.00	50	23.823	0.744	-1.240	4.346	4.162	8.369	292993.349	1.622
5229	28	31	15.00	10	12.332	3.417	0.320	3.961	3.564	6.028	14.707	2.880
5261	45	43	10.00	30	26.183	0.965	-1.578	4.361	4.651	2.525	83664.604	0.965
5357	36	22	5.00	30	22.768	0.774	1.093	4.028	4.506	3.022	8980.088	0.354
5390	22	17	4.00	30	17.724	0.819	0.513	3.650	4.317	4.262	1165.060	1.086
5497	44	24	4.00	30	25.292	0.652	0.642	4.231	4.571	3.005	9784887.950	0.233
6322	23	26	10.00	66	9.106	2.653	-0.405	3.753	3.536	5.852	48.308	3.981
6927	19	16	4.00	1	18.477	0.918	-0.601	3.925	4.247	4.178	8317.660	0.272
7115	19	39	4.00	30	15.827	0.848	0.444	4.314	3.226	3.401	408806.642	1.469
7162	51	14	1.00	50	20.424	0.808	-1.269	1.209	4.010	8.932	39471.599	1.093
7634	52	44	2.00	2	23.259	0.568	0.761	3.767	4.307	8.242	7635.257	1.509
7878	5	7	4.00	29	9.554	1.484	-0.612	3.815	3.285	1.450	619.191	0.875

8076	34	29	6.00	10	22.880	0.806	-1.387	4.321	4.419	2.903	8518.348	1.134
8078	29	32	4.00	36	8.066	1.796	1.664	3.587	3.088	10.469	143.391	5.352
8219	16	23	0.40	22	15.559	0.604	-0.087	2.601	1.940	14.337	674373.206	1.542
8396	53	59	0.50	1	13.606	1.385	7.016	0.880	0.767	16.517	1158.201	2.451
9120	52	44	10.00	20	28.586	0.876	-0.006	4.690	4.746	1.789	49444.336	0.219
9949	21	35	5.00	20	21.022	0.923	1.016	4.539	3.565	3.200	14679.381	0.563
10200	22	25	4.00	9	15.905	1.513	-7.127	3.619	3.583	6.466	28779.845	1.034
10684	16	58	4.00	57	18.929	0.659	-7.058	4.450	3.642	4.785	32.622	1.342
13333	18	19	2.00	25	19.948	0.638	0.168	3.975	3.913	6.246	85026.315	0.079
14242	68	37	3.00	20	31.135	0.558	0.383	4.266	4.663	5.096	2915.546	0.372
14420	56	20	20.00	22	13.393	1.571	-3.492	3.144	4.609	1.464	5.884	5.899
18045	40	50	4.00	30	26.932	0.654	-3.178	4.269	4.344	3.532	187.042	0.809
19018	63	67	8.00	30	29.425	0.835	1.891	4.634	4.360	2.918	18544.514	0.863
19991	22	50	4.00	20	24.460	0.658	0.062	4.599	3.990	3.472	925.409	1.058
20270	20	16	4.00	30	18.848	0.815	-0.255	3.707	4.250	3.938	293.230	0.531
21669	38	37	1.00	21	17.621	0.831	-1.649	1.917	2.520	14.959	2382.409	2.604
22962	23	18	4.00	24	17.168	0.879	-1.295	3.760	4.158	4.583	7906.999	1.143
23249	21	40	1.00	16	24.123	0.532	0.538	2.978	2.075	7.226	1795533.078	0.101
23652	24	24	1.00	20	20.607	0.643	-0.193	2.862	3.456	14.124	87142.434	1.056
37077	11	46	4.00	24	19.329	0.886	0.584	4.311	2.254	3.507	10417.255	1.563
38128	27	52	8.00	42	25.657	0.808	-0.562	4.687	4.444	1.655	23943.751	0.263
38760	10	17	5.00	33	12.412	1.155	0.026	4.192	3.255	2.936	6387644.371	2.551

Acknowledgments
The authors gratefully acknowledge the valuable research assistance provided by Simian Zhang during the course of this study.

References
Ales L, Cho SH, Körpeoğlu E (2017) Optimal award scheme in innovation tournaments. Operations Research 65(3):693–702, URL http://dx.doi.org/10.1287/opre.2016.1575.

Ales L, Cho SH, Körpeoğlu E (2021) Innovation tournaments with multiple contributors. Production and Operations Management 30(6):1772–1784, URL http://dx.doi.org/10.1111/poms.13342.

Aoyagi M (2010) Information feedback in a dynamic tournament. Games and Economic Behavior 70(2):242–260, URL http://dx.doi.org/10.1016/j.geb.2010.01.013.

Barrau A, Bonnabel S (2017) The invariant extended kalman filter as a stable observer. IEEE Transactions on Automatic Control 62(4):1797–1812, ISSN 0018-9286, 1558-2523, URL http://dx.doi.org/10.1109/TAC.2016.2594085.

Bensoussan A (1992) Stochastic Control of Partially Observed Systems (Cambridge University Press), 1 edition, ISBN 9780511526503, URL http://dx.doi.org/10.1017/CBO9780511526503.

Bergemann D, Heumann T, Morris S, Sorokin C, Winter E (2022) Optimal information disclosure in classic auctions. American Economic Review: Insights 4(3):371–388, ISSN 2640-205X, 2640-2068, URL http://dx.doi.org/10.1257/aeri.20210504.

Betancourt M (2017) A conceptual introduction to hamiltonian monte carlo. URL http://dx.doi.org/10.48550/ARXIV.1701.02434, version Number: 2.

Bimpikis K, Ehsani S, Mostagir M (2019) Designing dynamic contests. Operations Research 67(2):339–356, ISSN 0030-364X, 1526-5463, URL http://dx.doi.org/10.1287/opre.2018.1823.

Budd C, Harris C, Vickers J (1993) A model of the evolution of duopoly: Does the asymmetry between firms tend to increase or decrease? The Review of Economic Studies 60(3):543–573, URL http://dx.doi.org/10.2307/2298124.

Chen P, Pavlou P, Wu S, Yang Y (2021) Attracting high-quality contestants to contest in the context of crowdsourcing contest platform. Production and Operations Management 30(6):1751–1771, ISSN 1059-1478, URL http://dx.doi.org/10.1111/poms.13340.

Frogerais P, Bellanger JJ, Senhadji L (2012) Various ways to compute the continuous-discrete extended kalman filter. IEEE Transactions on Automatic Control 57(4):1000–1004, ISSN 0018-9286, 1558-2523, URL http://dx.doi.org/10.1109/TAC.2011.2168129.

Guerre E, Perrigne I, Vuong Q (2000) Optimal nonparametric estimation of first-price auctions. Econometrica 68(3):525–574, ISSN 0012-9682, 1468-0262, URL http://dx.doi.org/10.1111/1468-0262.00123.

Harris C, Vickers J (1987) Racing with uncertainty. The Review of Economic Studies 54(1):1–21, ISSN 00346527, 1467937X, URL http://www.jstor.org/stable/2297442.

Hoffman MD, Gelman A (2014) The no-u-turn sampler: Adaptively setting path lengths in hamiltonian monte carlo. Journal of Machine Learning Research 15(47):1593–1623, URL http://jmlr.org/papers/v15/hoffman14a.html.

Huang Y, He M (2021) Structural analysis of tullock contests with an application to U.S. house of representatives elections. International Economic Review 62(3):1011–1054, ISSN 1468-2354, URL http://dx.doi.org/10.1111/iere.12502.

Hwang SH (2012) Technology of military conflict, military spending, and war. Journal of Public Economics 96(1-2):226–236, ISSN 00472727, URL http://dx.doi.org/10.1016/j.jpubeco.2011.09.003.

Jiang ZZ, Huang Y, Beil DR (2022) The role of feedback in dynamic crowdsourcing contests: A structural empirical analysis. Management Science 68(7):4858–4877, ISSN 0025-1909, 1526-5501, URL http://dx.doi.org/10.1287/mnsc.2021.4140.

Kang K (2016) Policy influence and private returns from lobbying in the energy sector. The Review of Economic Studies 83(1):269–305, ISSN 0034-6527, 1467-937X, URL http://dx.doi.org/10.1093/restud/rdv029.

Khorasani S, Nittala L, Krishnan VV (2023) Screening in multistage contests. Manufacturing & Service Operations Management msom.2021.0378, ISSN 1523-4614, 1526-5498, URL http://dx.doi.org/10.1287/msom.2021.0378.

Lemus J, Marshall G (2021) Dynamic tournament design: Evidence from prediction contests. Journal of Political Economy 129(2):383–420, ISSN 0022-3808, URL http://dx.doi.org/10.1086/711762.

Lewis PAW, Shedler GS (1979) Simulation of nonhomogeneous poisson processes by thinning. Naval Research Logistics Quarterly 26(3):403–413, ISSN 0028-1441, 1931-9193, URL http://dx.doi.org/10.1002/nav.3800260304.

Mihm J, Schlapp J (2019) Sourcing innovation: On feedback in contests. Management science 65(2):559–576, URL http://dx.doi.org/10.1287/mnsc.2017.2955.

Moldovanu B, Sela A (2001) The Optimal Allocation of Prizes in Contests. American Economic Review 91(3):542–558, ISSN 0002-8282, URL http://dx.doi.org/10.1257/aer.91.3.542.

Moscarini G, Smith L (2011) Optimal dynamic contests, URL https://campuspress.yale.edu/moscarini/files/2017/01/contests_current-x5l31k.pdf, working paper.

Neal RM (1996) Bayesian Learning for Neural Networks. Lecture Notes in Statistics (New York, NY: Springer New York), ISBN 978-0-387-94724-2 978-1-4612-0745-0, URL http://dx.doi.org/10.1007/978-1-4612-0745-0, iSSN: 0930-0325.

Neal RM (2011) Mcmc using hamiltonian dynamics. Brooks S, Gelman A, Jones G, Meng XL, eds., Handbook of Markov Chain Monte Carlo, 113–162 (Chapman & Hall / CRC Press), URL http://dx.doi.org/10.1201/b10905, chapter 5.

Risdal M, Bozsolik T (2022) Meta kaggle. URL http://dx.doi.org/10.34740/KAGGLE/DS/9.

Ryvkin D (2022) To fight or to give up? dynamic contests with a deadline. Management Science 68(11):8144–8165, ISSN 0025-1909, 1526-5501, URL http://dx.doi.org/10.1287/mnsc.2021.4206.

Shakhgildyan K (2022) Nonparametric identification and estimation of all-pay auction and contest models. Review of Economic Design ISSN 1434-4750, URL http://dx.doi.org/10.1007/s10058-022-00309-3.

Skaperdas S (1996) Contest success functions. Economic Theory 7(2):283–290, ISSN 0938-2259, 1432-0479, URL http://dx.doi.org/10.1007/BF01213906.

Yoganarasimhan H (2016) Estimation of beauty contest auctions. Marketing Science 35(1):27–54, ISSN 0732-2399, URL http://dx.doi.org/10.1287/mksc.2015.0929.

Zhang S, Singh PV, Ghose A (2019) A structural analysis of the role of superstars in crowdsourcing contests. Information Systems Research 30(1):15–33, ISSN 1047-7047, URL http://dx.doi.org/10.1287/isre.2017.0767.